

Chern-Simons, CMI. *

Abstract

The notes-in-progress around Chern-Simons theory.

Contents

1	Classical Chern-Simons action	2
1.1	Recap on connections	2
1.2	Chern-Weil theory	4
1.3	Chern-Simons action	5
1.4	Gauge invariance of Chern-Simons	6
1.5	Equivariant connections	8
1.6	Projectively flat connections	10
1.7	Twisted differential operators	11
1.8	Modules over TDO's and projectively flat connections	14
1.9	Moduli of bundles	15
2	Conformal Field Theory	15
3	Quantization of Chern-Simons	15
3.1	Path integrals in a nutshell	16
3.2	Example: abelian Chern-Simons over S^3 and linking number of knots	18
3.3	Decorated cobordisms and Wilson observables	19
3.4	Canonical quantization	22
3.5	Jones polynomial	22
3.6	Modular functor and TQFT	22
3.7	S-matrix and Verlinde's coupling constants in Chern-Simons	22
3.8	Demystification of Verlinde's formula	24
3.9	Verlinde's algebra in unitary TQFT	25
3.10	Dimension of conformal blocks and S-matrix	27
4	Abelian varieties	29
4.1	Line bundles on abelian varieties	29
4.2	Correspondences and Poincaré line bundle	31
4.3	Some algebraic-geometry	33
4.4	Ample line bundles, positivity and polarizations	35
4.5	Principal polarizations and Siegel's upper half-space	37
5	Deformations and the geometric quantization	39
5.1	Deformations of complex structures	39
5.2	Deformation of Kähler polarization after Hitchin	41

*Mistakes are to be sent to Alexander Zakharov, zakharov@cml.ac.in; Date 14.06.26

1 Classical Chern-Simons action

1.1 Recap on connections

Let M be a manifold. Instead of working in the general setup of principal bundles we prefer to consider vector bundles equipped with some extra geometric structure. For example the following categories (groupoids to be precise) are equivalent:

1. $GL_n(\mathbb{R})$ -bundles and $\text{rk} = n$ real vector bundles.
2. $GL_n(\mathbb{C})$ -bundles and $\text{rk} = n$ complex vector bundles.
3. $U_n(\mathbb{C})$ -bundles and $\text{rk} = n$ complex vector bundles with hermitian metric.
4. $SU_n(\mathbb{C})$ -bundles and $\text{rk} = n$ complex vector bundles E with hermitian metric h and a section $s \in \Lambda^n E$ such that $h(s, s) = 1$.

We denote by $\Lambda_M^1 = T_M^*$ the cotangent bundle and $\Lambda_M^i = \Lambda^i(T_M^*)$ the bundle of differential i -forms. The tangent bundle T_M will be denoted by Λ_M^{-1} as well.

Assume E is a complex vector bundle. We will denote by $E(U)$ the set of sections of E over an open $U \subset M$. It is convenient to think of E in terms of its sections, so we will omit U from the notation.

The following notion allows to speak about derivatives of sections of E along tangent vectors T_M .

Definition 1.1.1. A *connection* ∇ on E is a map

$$\nabla: E(U) \rightarrow \Lambda^1 \otimes_{\mathbb{R}} E(U)$$

satisfying the Leybniz rule $\nabla_v(fs) = v(f)s + f\nabla_v(s)$ for all $v \in T_M$ and $f \in \Lambda_M^0$.

Remark 1.1.2. It is important that the connection on E automatically provides a connection on any natural bundle associated with E : E^* , $\text{End}(E)$, $\Lambda^i E$, etc. This corresponds to the statement that any Lie algebra representation acts on its tensors. For example ∇ acts on the dual E^* , $\text{End}(E) \simeq E^* \otimes E$, etc. The rule of thumb is that tensor products and natural pairings are with respect to the connections. For instance if $\phi \in \text{End}(E)(U)$, then $(\nabla\phi)(s) = \nabla(\phi(s)) - \phi(\nabla s)$.

Note that the difference of any two connections $\nabla' - \nabla$ is Λ_M^0 -linear, i.e. $\nabla' - \nabla$ is equal to some $A \in \Lambda_M^1 \otimes \text{End}(E)$. Thus the space

$$\mathcal{C}(E) = \{\text{a connection } \nabla \text{ on } E\}$$

is an *affine space* over $\Lambda_M^1 \otimes \text{End}(E)(M)$: the latter acts via $\nabla \rightsquigarrow \nabla + A$. In particular $T_{\nabla \in \mathcal{C}} \simeq \Lambda_M^1 \otimes \text{End}(E)(M)$ for all ∇ .

Locally E is trivial, after fixing basis sections we get an identification $E \simeq \oplus_{i \leq k} \Lambda_M^0$: any connection (locally) has the form $\nabla = d + A$, where d is $\Lambda_M^0 \rightarrow \Lambda_M^1$ and $A \in \Lambda_M^1 \otimes \text{End}(E)$.

Exercise Assume M is a line bundle with connection ∇ . Suppose L is N -th root of M , i.e. an isomorphism $L^{\otimes N} \simeq M$ is fixed. Show that L admits a natural connection induced by ∇ as follows. Choose non vanishing $s \in L(U)$, then $\nabla(s^{\otimes N}) = \nu \cdot s^{\otimes N}$ for some $\nu \in \Lambda^1$. Set $\nabla^{1/N}(s) = \frac{1}{N}\nu \cdot s$. Show that $\nabla^{1/N}$ doesn't depend on s and defines a connection on L .

Exercise Given the connection ∇ and any smooth path $\gamma: [0, 1] \rightarrow M$, for each $t \in [0, 1]$ there is a linear map $\Pi_t: E_{\gamma(0)} \rightarrow E_{\gamma(t)}$ called the *parallel transport* along γ . Define a connection

$\gamma^*(\nabla)$ on $\gamma^*(E)$ over the interval and show that for each $v \in \gamma^*(E)_0$ there is a unique section $\tilde{v} \in \gamma^*(E)([0, 1])$ such that $\gamma^*(\nabla)\tilde{v} = 0$. By the definition $\Pi_t(v) = \tilde{v}_t$.

Assume that in some gauge $\nabla = d + A$, then $\gamma^*(\nabla) = A(\dot{\gamma})dt$. Check that the matrix of $\Pi_{\nabla, \gamma}$ in the gauge is equal to the time-ordered exponent $\text{Pexp}(-A(\dot{\gamma}))$, given by the iterated integral formula

$$\text{Pexp}(X(t)) = \text{id} + \int_0^t X(t_0)dt_0 + \int_0^t X(t_0)dt_0 \int_0^{t_0} X(t_1)dt_1 + \dots$$

satisfies $\frac{d}{dt} \text{Pexp}(X(t)) = \dot{X} \text{Pexp}(X(t))$.

The gauge group $\text{Aut}(E)$ acts on $\Lambda_M^1 \otimes \text{End}(E)$ and $\mathcal{C}(E)$ differently. For $g \in \text{Aut}(E)$ we have $g.\nabla = g \circ \nabla \circ g^{-1}$. If $\nabla = d + A$, then $g(d + A)g^{-1} = d + gdg^{-1} + gAg^{-1}$. Roughly speaking, the transformation of a 1-form A , viewed as a *connection*, acquire the term gdg^{-1} .

Exercise Fix a Lie group G (e.g. $\text{GL}_n(\mathbb{C})$) with the Lie algebra $\mathfrak{g} \simeq T_{e \in G}$. The gauge transformation g is a map $g: M \rightarrow G$. Check that

$$-gdg^{-1} = g^{-1}dg = g^*(\mu),$$

where $\mu \in \Lambda_G^1 \otimes \mathfrak{g}$ is the Maurer-Cartan form, i.e. the unique left invariant form such that $\mu_e(v) = v$ for any $v \in T_{e \in G}$. Show that μ satisfy the Maurer-Cartan equation:

$$d\mu + \frac{1}{2}[\mu, \mu] = d\mu + \mu \wedge \mu = 0.$$

Recall that the de Rham differential

$$d: \Lambda_M^i(U) \rightarrow \Lambda_M^{i+1}(U)$$

is the unique extension of the map $\Lambda_M^0(U) \rightarrow \Lambda_M^1(U)$ which satisfies the super-commutative Leibniz rule:

$$d(a \wedge b) = da \wedge b + (-1)^{|a|} a \wedge db.$$

Definition 1.1.3. The *curvature* of the connection $\nabla = d + A$ is 2-form

$$F^\nabla = d(A) + A \wedge A: \Lambda_M^2 \otimes \text{End}(E).$$

Exercise Check that 2-form F^∇ is independent of coordiantes: $F^{g.\nabla} = gF^\nabla g^{-1}$. Check that the following formula makes sense

$$F^\nabla = \frac{1}{2}[\nabla, \nabla] \in E \rightarrow \Lambda_M^2 \otimes E.$$

Deduce Bianchi's identity:

$$\nabla(F^\nabla) = [\nabla, [\nabla, \nabla]] = 0 \in \Lambda_M^3 \otimes \text{End}(E).$$

By the definition, a hermitian form $h(-, -)$ on E satisfy $h(ta, b) = h(a, \bar{t}b) = th(a, b)$ and $h(a, b) = \overline{h(b, a)}$ for all $a, b \in E(M)$ and $t \in \mathbb{C}$.

Exercise Show that any complex vector bundle E admits a hermitian form $h(-, -)$.

The $U_n(\mathbb{C})$ -bundle corresponding to E and an hermitian form $h(-, -)$ is denoted by (E, h) . Similarly, $SU_n(\mathbb{C})$ -bundle is a triple (E, h, s) , where $s \in \Lambda_{\mathbb{C}}^n E$ such that $h(s, s) = 1$.

Exercise Check that the complex vector bundle E lifts to a $SU_n(\mathbb{C})$ -bundle, if and only if, $c_1(E) = 0 \in H^2(M; \mathbb{Z})$.¹

Let $\mathcal{C}(E, h) \subset \mathcal{C}(E)$ denotes the set of all *unitary* connections ∇ satisfying

$$dh(a, b) = h(\nabla a, b) + h(a, \nabla b).$$

Similarly $\mathcal{C}(E, h, s) \subset \mathcal{C}(E, h)$ we be the set of *special unitary* connections ∇ satisfying $\nabla(s) = 0$.

One can see that $\mathcal{C}(E, h)$ is an affine space over $\Lambda_M^1 \otimes \mathfrak{u}(E, h)$ and $\mathcal{C}(E, h, s)$ is an affine space over $\Lambda_M^1 \otimes \mathfrak{su}(E, h, s)$, where $\mathfrak{u}(E, h)$ is the Lie algebra of the unitary group $\text{Aut}(E, h)$ and $\mathfrak{su}(E, h, s)$ is the Lie algebra of the special unitary group $\text{Aut}(E, h, s)$. Note that the curvature F^∇ always takes values in the corresponding Lie algebras.

1.2 Chern-Weil theory

Assume E is a complex vector bundle over some manifold M . Suppose that ∇ is a connection on E and $F(\nabla)$ its curvature.

Exercise Consider variation $\nabla \rightsquigarrow \nabla + \delta A$. Check that $\delta F(\nabla) = \nabla(\delta A)$. Use Bianchi identity to show that $d \text{tr}(F(\nabla)^k) = 0$ and

$$\delta \text{tr}(F(\nabla)^k) = k \cdot d(\text{tr}(F(\nabla)^{k-1} \delta A))$$

for all k . Deduce that $[\text{tr}(F(\nabla)^k)] \in H^{2k}(M; \mathbb{C})$ is well-defined and independent of ∇ .

Recall that Newton's polynomials $\sum_{k \geq 1} t^k \text{tr}(F^k)$ are expressed in terms of characteristic polynomial $\det(\text{id} + tF)$, and vice versa.

Corollary 1.2.1. *We have*

$$[\det(\text{id} - \frac{1}{2\pi i} F(\nabla))] = c(E) \in H^*(M; \mathbb{Z}),$$

where $c(E) = 1 + c_1(E) + c_2(E) + \dots$, is the Chern polynomial.

Proof. Both parts are multiplicative in E . Use the splitting principle to reduce to the case E is a line bundle. The classifying space for line bundles is $\mathbb{C}P^\infty$ with the universal line bundle $\mathcal{O}(-1)$. On the other hand $c_1(\mathcal{O}(-1))$ is detected by its restriction to the submanifold $\mathbb{C}P^1$. The only issue is to determine the coefficient before F in the case of the line bundle $\mathcal{O}(-1)$ with, say, Chern connection over $\mathbb{C}P^1$.

In the restriction to $\mathbb{C} \subset \mathbb{C}P^1$, the curvature of the Chern connection on $\mathcal{O}(-1)$ is equal to $F = \frac{d\bar{z}dz}{(1+|z|^2)^2}$ (in the restriction to $\mathbb{C} \subset \mathbb{C}P^1$). Hence $\int_{\mathbb{C}P^1} F = 2\pi i$, on the other hand $c_1(\mathcal{O}(-1)) = -1$, so $c_1 = -\frac{1}{2\pi i}[F]$. $\square$

¹There are many ways to define $c_1(E)$, see below. You can use the fact that $c_1(E) = 0 \iff \det E = \Lambda^{\text{rk } E} E$ admits smooth non vanishing section.

1.3 Chern-Simons action

Let M be a compact manifold of $\dim M = 3$ with possibly non empty boundary $\Sigma = \partial M$. We assume that M is oriented, i.e. one can define the integral $\int_M v_3$ for any top-form $v_3 \in \Lambda^3(M)$.

It turns out that the above section considerations simplify because $SU(N)$ -bundles over M and ∂M are in fact trivial.

Proposition 1.3.1. *Any $SU(N)$ -bundle E on M is trivial.*

Proof. Recall that the isomorphism classes of $SU(N)$ -bundles on M are in 1-1 correspondence with the set of maps $[M, BSU(N)]$ up to homotopy. It is well-known that $\pi_{<3}(SU(N)) = 0$, hence $\pi_{<4}(BSU(N)) = 0$ and so $[M, BSU(N)] = 0$ for any 3-dimensional cellular complex M . Thus E is isomorphic to the trivial one. $\square$

To complete the discussion we sketch the following topological result.

Theorem 1.3.2 (Rokhlin). *The tangent bundle T_M is trivial.*

Proof. Passing to the double construction $M \cup_{\partial M} \bar{M}$, one can assume that $\partial M = \emptyset$. here we abuse notation and denote by $[M]$ the reduction $[M] \in H^3(M; \mathbb{Z}/2)$ modulo 2.

Denote by $w_i \in H^i(M; \mathbb{Z}/2)$, $i = 1, 2, 3$ the Stiefel-Whitney classes of T_M . Recall that $w_1 = 0$ iff M is orientable and $w_3 = \chi(M) \cdot [M]$ is the Euler characteristics times the fundamental class of M modulo 2. Hence $w_1, w_2 = 0$.

1. Let $w = 1 + w_1 + w_2 + w_3 \in H^*(M; \mathbb{Z}/2)$. For any $x \in H^*(M; \mathbb{Z}/2)$ we denote by $\int_{[M]} x \in \mathbb{Z}/2$ the pairing of x with $[M] \in H^3(M; \mathbb{Z}/2)$. Recall the Wu's formula:

$$\int_{[M]} \text{Sq}(x) = \int_{[M]} x \text{Sq}^{-1}(w),$$

where $\text{Sq}(x) = \sum_{i \geq 0} \text{Sq}^i(x)$ is the automorphism of $H^*(M; \mathbb{Z}/2)$ given by the full Steenrod square and Sq^{-1} is its inverse. In our case $\text{Sq}^{-1}(w) = \text{Sq}^{-1}(1 + w_2) = 1 + w_2$, hence

$$\int_{[M]} w_2 \cup x = \int_{[M]} \text{Sq}^2(x_1)$$

for any $x_1 \in H^1(M; \mathbb{Z}/2)$. Since $\text{Sq}^{>1}(x_1) = 0$ and Poincaré pairing is not degenerate we conclude that $w_2 = 0$.

2. It is well-known that $w_2 = 0$ implies that T_M admits a spin structure. Recall that we have a double covering $SU(2) \simeq \text{Spin}(3) \rightarrow \text{SO}(3)$ and the spin structure means T_M is a reduction of some $SU(2)$ -bundle such that its reduction to $\text{SO}(3)$ is equivalent (as real vector bundle) to T_M . By the above any $SU(2)$ -bundle on M is trivial, hence T_M is trivial as well. $\square$

From general cobordism theory it follows that any oriented M with $\partial M = \emptyset$ is the boundary of a 4-dimensional oriented manifold N . Since E is trivial, one can extend it, as well as ∇ , to a connection $\tilde{\nabla}$ on a trivial bundle over N .

In the following we will work differential forms with values in $\text{Mat}_{n \times n}(\mathbb{C})$. Recall that tr is symmetric form on matrices, hence for any collection of forms $a_i \in \Lambda^{d_i} \otimes \text{Mat}_{n,n}$, $i \leq k$ one can define a form $\text{tr}(a_1 \wedge \dots \wedge a_k) \in \Lambda^d$, where $d = \sum_i d_i$. The usual symmetries of tr work in this graded commutative setting as well.

Exercise Check that $\text{tr}(a_1 \wedge \dots \wedge a_k) = (-1)^{d'} \text{tr}(a_2 \wedge a_3 \wedge \dots \wedge a_k \wedge a_1)$, $d' = d_1(d_2 + d_3 + \dots + d_k)$ is equal to the number of “moves” of a_1 through $a_2, \dots, a_k$.

Exercise Show that for any $SU(n)$ -bundle $\tilde{E}$ on 4-dimensional N with a connection $\tilde{\nabla}$ and its curvature F , we have

$$\frac{1}{4\pi^2} \int_N \text{tr}(F \wedge F) = \int_{[N]} c_2(E).$$

If $\partial N = M$, then

$$\frac{1}{4\pi^2} \int_M \text{tr}(F(\tilde{\nabla}) \wedge F(\tilde{\nabla})) \in \mathbb{R}/\mathbb{Z}$$

is well-defined and is independent of N and the extension $\tilde{\nabla}$.

If $\tilde{\nabla} = d + \tilde{A}$ in some trivialization of $\tilde{E}$, then one immediately checks that

$$d \text{tr}(\tilde{A} \wedge d\tilde{A} + \frac{2}{3} \tilde{A} \wedge \tilde{A} \wedge \tilde{A}) = \text{tr}(F(\tilde{\nabla}) \wedge F(\tilde{\nabla})).$$

Restriction of the trivialization to the boundary allows to write $\nabla = d + A$, we obtain that

$$\frac{1}{8\pi^2} \int_M \text{tr}(AdA + \frac{2}{3} AAA) \in \mathbb{R}/\mathbb{Z},$$

is a well-defined invariant of M and A .

That the way how Chern and Simons came up to this functional. To be consistent with Witten's notation we define Chern-Simons functional by

$$S_{CS}(A) = \frac{1}{4\pi} \int_M \text{tr}(AdA + \frac{2}{3} AAA) \in \mathbb{R}/2\pi\mathbb{Z}.$$

Thus, for closed M , $e^{ikS_{CS}(A)}$ is gauge invariant for any integer k . For convenience we will set $k' = \frac{k}{4\pi}$ and $S'_{CS}(A) = \int_M \text{tr}(AdA + \frac{2}{3} AAA)$.

1.4 Gauge invariance of Chern-Simons

Here we will consider the case of 3-manifold M with boundary. Let $\mathcal{C} = \mathcal{C}(E, h, s)$ denotes the space of special unitary connections on trivial bundle E . Under the variation $A \rightsquigarrow A + \delta A$, we have:

$$\delta S'_{CS} = \int_{\partial M} \text{tr}(\delta A \wedge A) + 2 \int_M \text{tr}(F(\nabla) \delta A) = \nu_{\nabla}(\delta A) + 2\eta_{\nabla}(\delta A).$$

The formula defines two 1-forms $\nu, \eta \in \Lambda^1_{\mathcal{C}}$. Since ν depends only on the restriction to ∂M , we will consider the corresponding term $\nu^B \in \Lambda^1_{\mathcal{C}(\partial M)}$ as well.

If we go beyond 0-forms and denote the de Rham differential on the space $\mathcal{C}(M)$ by $\mathcal{D}$, then the above equality reads as $\mathcal{D}(\nu + 2\eta) = 0$. Using the usual formula for $\mathcal{D}$ we obtain that

$$\mathcal{D}\nu^B(\delta A, \delta B) = - \int_{\partial M} \text{tr}(\delta A \wedge \delta B),$$

is equal to the symplectic form on $\mathcal{C}(\partial M, E, h)$, it is affine parallel.

The form η is clearly gauge invariant, while ν is not. Namely, since $g \cdot \delta A = g \delta A g^{-1}$ and $g \cdot A = g A g^{-1} + g dg^{-1}$ we obtain

$$(g^* \nu - \nu)(\delta A) = \int_{\partial M} \text{tr}(g^{-1} dg \wedge \delta A) = \delta f_g,$$

where the functional $f_g(-) \in \Lambda_{\mathcal{C}}^0$ is given by

$$f_g(A) = \int_{\partial M} \text{tr}(g^{-1}dg \cdot A). \quad (1.4.1)$$

We obtain

$$\delta(g^*S' - S') = (g^*\nu - \nu)(\delta A) = \delta f_g.$$

Hence for any A we have $S'(g.A) - S'(A) - f_g(A) = C_g$ for some C_g which depends only on g . To recover C_g it is enough to compare both parts at $A = 0$; this simple computation shows an equality for all A :

$$S'(g.A) - S'(A) = \int_{\partial M} \text{tr}(g^{-1}dg \wedge A) + \frac{1}{3} \int_M \text{tr}(g^{-1}dg \wedge g^{-1}dg \wedge g^{-1}dg). \quad (1.4.2)$$

Let us take a closer look at the terms appearing on the right hand side.

1. The second term in (1.4.2):

$$S'_{top}(M, g) = \frac{1}{3} \int_M (g^{-1}dg \wedge g^{-1}dg \wedge g^{-1}dg),$$

appears in the Witten's correction to Wess-Zumino sigma-model on ∂M and up to a normalization its exponent is determined by a restriction to ∂M . As we know $g^{-1}dg = g^*(\mu)$, where μ is the Maurer-Cartan form on $\text{SU}(N)$. It is known that $\int_{\text{SU}(2)} \text{tr}(\mu \wedge \mu \wedge \mu) = 24\pi^2$ and hence the biinvariant form $w_3 = \frac{1}{24\pi^2} \text{tr}(\mu \wedge \mu \wedge \mu) \in \Lambda^3(\text{SU}(N))$ represents the generator of $H^3(\text{SU}(N); \mathbb{Z})$. In particular, for closed M one has

$$S'_{top}(M, g) = 8\pi^2 \cdot \deg(g),$$

where $\deg(g) = \int_M w_3 \in \mathbb{Z}$. In the case of gauge group $\text{SU}(2)$, $\deg(g)$ is the degree of the map $g: M \rightarrow \text{SU}(2) = S^3$. This confirms that $e^{ik'S'_{CS}}, k \in \mathbb{Z}$ is gauge invariant if $\partial M = \emptyset$.

Lets go back to the general case $\partial M \neq \emptyset$. It follows that the value $e^{i\frac{k}{4\pi}S'_{top}(M, g)}$ depends only on the restriction $g|_{\partial M}: \partial M \rightarrow \text{SU}(N)$.

A direct calculations shows that for any $g_1, g_2: M \rightarrow \text{SU}(N)$ we have Polyakov–Wiegmann formula

$$S'_{top}(M, g_1 g_2) - S'_{top}(M, g_1) - S'_{top}(M, g_2) = -\text{PW}(g_1, g_2),$$

where we set

$$\text{PW}(g_1, g_2) = \int_{\partial M} \text{tr}(g_1^{-1}dg_1 \wedge dg_2 g_2^{-1}).$$

2. The term $f_g(A)$ depends only on a restriction to ∂M and a simple calculation shows that

$$f_{g_1 g_2}(A) - f_{g_1}(g_2.A) - f_{g_2}(A) = \text{PW}(g_1, g_2).$$

The discussion above implies that

$$e_g(A) = e^{\frac{i}{4\pi}(f_g(A) + S'_{top}(g))}$$

depends only on the restrictions to ∂M and satisfies the *cocycle condition*:

$$e_{g_1 g_2}(A) = e_{g_1}(g_2.A) \cdot e_{g_2}(A),$$

moreover we have

$$e^{ik_{SCS}(g.A)} = e^{ik_{SCS}(A)} \cdot e_g^k(A). \quad (1.4.3)$$

In other words $e_g(A)$ defines an equivariant structure on the trivial line bundle $\mathcal{C}(M, h) \times \mathbb{C}$ with respect to the action of $\text{Maps}(M, \text{SU}(N))$ via the identification of fibers at A and $g.A$, respectively:

$$e_g(A): \mathbb{C}_A \rightarrow \mathbb{C}_{g.A}.$$

The cocycle $e_g(A)$ defines $\text{U}(1)$ -bundle $\tilde{\mathcal{L}}$ on the quotient $\mathcal{C}(M)/\text{Maps}(M, \text{SU}(N))$. Since $e_g(A)$ depends only on the restriction of the connection A and the gauge g to the boundary ∂M , the same construction defines $\text{U}(1)$ -bundle $\mathcal{L}$ over $\mathcal{C}(\partial M)$ and the restriction map $p: \mathcal{C}(M) \rightarrow \mathcal{C}(\partial M)$ provides the identification $\tilde{\mathcal{L}} = p^*\mathcal{L}$.

Thus we can consider the function $A \rightsquigarrow e^{ik_{SCS}(A)}$, as a unitary section of a natural $\text{U}(1)$ -bundle over $\mathcal{C}(M, \text{SU}(N))$:

$$e^{ik_{SCS}} \in \Gamma(p^*\mathcal{L}^{\otimes k}), \quad (1.4.4)$$

4. The standard approach to a quantization of gauge theory via path integrals is to first eliminate the redundancy provided by the gauge action and make sense of the integration over the space of orbits. In other words one has to pass from $\mathcal{C}(M)$ and the action $e^{ik_{SCS}}$ with values in the trivial line bundle, to the quotient $\mathcal{C}(M)/\text{Maps}(M, \text{SU}(N))$ with $e^{ik_{SCS}}$ viewed a section of the line bundle $\mathcal{L}^{\otimes k}$. In 3.3 we will see that the Wilson observables admit a similar description.

1.5 Equivariant connections

Assume X is a manifold provided with an action of Lie group G . Suppose E is G -equivariant bundle over X . In other words for each $x \in X$ and $g \in G$ we have an isomorphism $e_g(x): E_x \rightarrow E_{g.x}$ satisfying the cocycle condition:

$$e_{g_1 g_2}(x) = e_{g_1}(g_2.x) e_{g_2}(x). \quad (1.5.1)$$

Special case is when $E = X \times V$ is a trivial bundle and the cocycle $e_g(x)$ is independent of x . In this case the cocycle condition on e is equivalent to the structure of G -representation on V . The corresponding bundle over the quotient is equal to

$$X \times V / (x, v) \sim (g.x, e_g \cdot v). \quad (1.5.2)$$

We would like to describe connections on E which come from X/G . Suppose ∇ is a connection on E . Then for any $g \in G$, the pullback of $g^*\nabla$ defines a connection on E via the identification $e_g: E \rightarrow g^*E$.

Proposition 1.5.3. *Suppose (E, ∇) locally takes the form $\nabla = d + \Phi$ for $\Phi \in \Lambda^1 \otimes \text{Mat}_{n \times n}(\mathbb{C})$, so that we can write $e_g(x) \in \text{GL}_n(\mathbb{C})$. Then*

$$g^*(\nabla) = d + g^*(e_g^{-1} de_g) + e_g^{-1} g^*(\Phi) e_g. \quad (1.5.4)$$

Proof. DIY. □

Remark 1.5.5. Here g^* denotes the usual pullback of differential forms. Note that this the action on the *right*. For the trivial action on X , $g^* = \text{id}$ and the formula recovers the usual gauge transformation formula, where we worked with the left action convention.

Definition 1.5.6. A connection ∇ on the equivariant bundle E is *invariant* if $g^*\nabla = \nabla$.

It is useful to have infinitesimal characterization of equivariance. Locally in the first order in δg viewed as an element of $\mathfrak{g}$ we can write

$$e_{1+\delta g}(x) = \text{id}_n + \mu_{\delta g} \in \text{GL}_n(\mathbb{C}).$$

In the standard notation this defines $\mu_a(x) \in \text{Mat}_{n \times n}(\mathbb{C})$ for any $a \in \mathfrak{g}$.

The element μ_a has no invariant meaning, because it's definition involves fibers over different points. Using ∇ one can set

$$\alpha_a = \mu_a + \Phi(X_a),$$

then it is easy to check the formula defines a tensor $\alpha: \mathfrak{g} \rightarrow \text{End}(E)$.

The variation of (1.5.4) with respect to $g = 1 + \delta g$, for $\delta g = a \in \mathfrak{g}$ gives:

$$X_a(\nabla) = d\mu_a + \mathcal{L}_{X_a}(\Phi) + [\Phi, \mu_a].$$

Substitution $\mu_a = \alpha_a - \Phi(X_a)$ provides an invariant expression

$$X_a(\nabla) = \nabla \alpha_a + X_a \lrcorner F(\nabla) \in \Lambda^1 \otimes \text{End}(E), \quad (1.5.7)$$

where $X_a \lrcorner$ is the contraction with the vector field X_a . The cocycle condition on e_g translates to

$$\mu_{[a,b]} = X_b \cdot \mu_a - X_a \cdot \mu_b + [\mu_a, \mu_b], \quad (1.5.8)$$

where X_a, X_b are vector fields on X induced by infinitesimal action of $a, b \in \mathfrak{g}$.

Remark 1.5.9. The cocycle condition on e_g is equivalent to the group homomorphism

$$f: G \rightarrow G \ltimes M,$$

which sends g to (g, e_g) . Here $M = \text{Maps}(X, \text{GL}_n(\mathbb{C}))$ and the right action by G via pullback turns $G \ltimes M$ into a semidirect product. On the level of Lie algebras we get a map $df: \mathfrak{g} \rightarrow \mathfrak{g} \ltimes \mathfrak{m}$. Here $\mathfrak{m}$ is the Lie algebra of M provided with the action of G , that will be written infinitesimally by $m \cdot a$ for $m \in \mathfrak{m}$ and $a \in \mathfrak{g}$. One can show that the Lie algebra structure on $\mathfrak{g} \ltimes \mathfrak{m}$ is given by the formula $[(a, m), (a', m')] = ([a, a'], [m, m'] + m \cdot a' - m' \cdot a)$. The infinitesimal cocycle condition says that df is a map of Lie algebras. More generally it is equivalent to the Maurer-Cartan equation for $\mu \in \Lambda^1 \mathfrak{g}^\vee \otimes \mathfrak{m}$ given by $d^{CE} \mu + \frac{1}{2}[\mu, \mu] = 0$, where d^{CE} is the Chevalley-Eilenberg differential.

This discussion also explains that *locally* in G in order to provide e_g satisfying (1.5.1) it is enough to find μ_a satisfying (1.5.8).

Invariantly we get

$$F(\nabla)(X_a, X_b) + \nabla_{X_b} \alpha_a - \nabla_{X_a} \alpha_b + [\alpha_a, \alpha_b] - \alpha_{[a,b]} = 0, \quad (1.5.10)$$

Remark 1.5.11. One can view α as an element in $\Lambda^* \mathfrak{g}^\vee[-1] \otimes \text{End}(E)$ (i.e. $\mathfrak{g}^\vee$ is in degree 1). The latter has natural degree 1 operators given by the Chevalley-Eilenberg differential d^{CE} acting on $\Lambda^* \mathfrak{g}^\vee$ and $m \circ \nabla$ induced by the composition

$$\text{End } E \rightarrow \Lambda^1 \otimes \text{End}(E) \rightarrow \mathfrak{g}^\vee \otimes \text{End } E,$$

where the last arrow is the Lie algebra action on X . Our equation is equivalent to the condition that $d^{CE} + m \circ \nabla + \alpha$ is a differential.

Assume s is an invariant section of E , i.e. $s(g \cdot x) = e_g(x)s(x)$. Infinitesimally this is the condition $X_a(s) = \mu_a s$, or in invariant terms

$$\nabla_{X_a} s = \alpha_a s. \quad (1.5.12)$$

TODO: Complex structure on $\mathcal{C}(\partial M, h, s)$ and the induced Chern connection of $\mathcal{L}$. Show that the curvature $F(\mathcal{L})$ is the pullback from GIT quotient.

1.6 Projectively flat connections

Fix a manifold X and a complex vector bundle E over X of rank N .

Definition 1.6.1. A connection ∇ on the complex vector bundle E is *projectively flat* if it has scalar curvature: $F(\nabla) = \omega \cdot \text{id}_E$, where $\omega \in \Lambda_M^2$.

Recall that there is a 1-1 correspondence between representations $\pi_1(X, x) \rightarrow \text{GL}_N(\mathbb{C})$ and flat connections. We would like to have a similar characterization of projectively flat connections. Clearly a projectively flat connection ∇ gives a flat connection on $\mathbb{P}(E)$. This is equivalent to a representation $\pi_1(X, x) \rightarrow \text{PGL}_n(\mathbb{C})$, the monodromy of $\text{PGL}(E)$ -connection, for some basepoint $x \in X$.

Exercise Suppose $\det(E)$ admits an N -th root bundle L , i.e. an isomorphism $L^{\otimes N} \simeq \det(E)$ is fixed. Then ∇ induces a natural connection on L compatible with ∇ . Check that if (E, ∇) is projectively flat, then the induced connection on $E \otimes L^{-1}$ is flat.

Let $p: X' \rightarrow X$ denotes the $\mathbb{C}^\times$ -bundle $\det(E)^\times = \det(E) \setminus \{0\}$. The pullback bundle $p^*\det(E)$ is provided with natural trivialization $s \in \Gamma(p^*\det(E))$ (namely $s_v = v$ for a vector $v \in \det(E)_x$ over x). Denote by L' the trivial bundle over X' equal to the “preferred” N -th root of $p^*(\det(E))$ over X' . By the above $p^*E \otimes L'^{-1} \simeq p^*E$ is provided with a natural *flat* connection.

To describe ∇' explicitly let $p^*(\nabla)(s) = \mu \cdot s$, then $p^*(F(\nabla))(s) = d\mu$ is equal to $N \cdot \omega$ and hence

$$\nabla' = p^*(\nabla) - \frac{\mu}{N} \cdot \text{id}_N, \quad (1.6.2)$$

is flat and in fact coincides with the description above. One checks that $\nabla'(s) = 0$. We proved the following.

Proposition 1.6.3. *For a projectively flat connection (E, ∇) there is a natural construction of a flat $\text{SL}_N(\mathbb{C})$ -bundle on $\det(E)^\times$.*

From the fiber sequence $S^1 \rightarrow T \rightarrow X$ we get an exact sequence

$$\pi_2(X) \rightarrow \pi_1(S^1) = \mathbb{Z} \rightarrow \pi_1(T) \rightarrow \pi_1(X) \rightarrow 1.$$

The homomorphism

$$\langle c_1(E), - \rangle: \pi_2(X) \rightarrow \mathbb{Z} = \pi_1(S^1),$$

evaluated on the spheroid $s: S^2 \rightarrow X$ is equal to the degree of $s^*(L)$.

(one can prove that the homomorphism is trivial iff L is a pullback under $X \rightarrow B\pi_1$.)

Denote its cokernel by $A = \mathbb{Z}/\langle c_1(L), \pi_2(X) \rangle$. In some interesting examples where $c_1(E) = 0$ or $\pi_2(X) = 0$ we have $A = \mathbb{Z}$. Hence any projectively flat connection (E, ∇) gives a central extension

$$0 \rightarrow A \rightarrow \tilde{\pi}_1 \rightarrow \pi_1 \rightarrow 1.$$

Recall that central extensions of a group π_1 by A are classified by the group cohomology $H^2(\pi_1; A) \simeq H^2(B\pi_1; A)$.

add some geometry?

1.7 Twisted differential operators

Fix a complex manifold X . We will follow [Gin98] and [BK+01].

For a line bundle L over X denote by $\text{Diff}_{\leq k}(L)$ the sheaf of differential operators of degree $\leq k$ acting on E . Recall that the associated graded of $\text{Diff}(L)$ is the commutative algebra $\text{Sym}(TX)$ and the isomorphism

$$\sigma: \text{Diff}_{\leq k}(L)/\text{Diff}_{< k} \simeq \text{Sym}^k(TX)$$

is called the symbol map.

Definition 1.7.1. Exact sequence of Lie algebras and $\mathcal{O}_X$ -modules

$$0 \rightarrow \mathcal{O} \xrightarrow{i} \mathcal{A} \xrightarrow{p=\sigma} TX \rightarrow 0, \quad (1.7.2)$$

is called *Atiyah's algebra* if the Lie bracket is compatible with the structure of $\mathcal{O}_X$ -modules: for any $D \in \mathcal{A}$ and $f \in \mathcal{O}_X$ we have

$$[D, i(1_{\mathcal{A}})] = 0$$

and

$$[D_1, fD_2] = f[D_1, D_2] + p(D_1) \cdot f \cdot D_2,$$

where $1_{\mathcal{A}} = i(1)$.

The basic example of Atiyah's algebra is $\text{Diff}_{\leq 1}(L)$. The trivial extension corresponds to ordinary differential operators of degree ≤ 1 . In analytic topology any $\mathcal{D}_{\leq 1}$ as above is locally trivial.

Denote by $T_{\mathbb{C}}(\mathcal{A})$ the sheaffication of the presheaf of free algebras generated by $\mathcal{A}$. The following is the appropriate version of universal enveloping algebra of the Lie algebra $\mathcal{A}$ over $\mathcal{O}$.

Definition 1.7.3. Given Atiyah's algebra $\mathcal{A}$ let $\mathcal{U}(\mathcal{A}/\mathcal{O})$ be the sheaffication of the quotient of $T_{\mathbb{C}}(\mathcal{A})$ by the relations

$$\begin{aligned} D_1 \otimes D_2 - D_2 \otimes D_1 &= [D_1, D_2], \\ f \otimes D &= fD, \quad f \in \mathcal{O}_X, D \in \mathcal{A}, \end{aligned}$$

and

$$1_{T_{\mathbb{C}}(\mathcal{A})} \simeq 1_{\mathcal{A}},$$

for all $f \in \mathcal{O}_X$ and $D_{1,2} \in \mathcal{A}$.

Definition 1.7.4. The sheaf of filtered algebras $\mathcal{U}(\mathcal{A}/\mathcal{O})$ is called the sheaf of *twisted differential operators* corresponding to Atiyah algebra $\mathcal{A}$.

The above definition is justified by the following observation.

Proposition 1.7.5. *There is natural isomorphism of filtered algebras*

$$\mathcal{U}(\text{Diff}_{\leq 1}(L)/\mathcal{O}) \simeq \text{Diff}(L).$$

Proof. The inclusion $\text{Diff}_{\leq 1}(L) \rightarrow \text{Diff}(L)$, induces a map of algebras $\mathcal{U}(\text{Diff}_{\leq 1}(L)/\mathcal{O}) \rightarrow \text{Diff}(L)$. It is well-known that locally it is an isomorphism. $\square$

It is common to consider the algebra of differential operators as a quantization of the Poisson algebra of functions $\mathcal{O}(T^*X)$ on the symplectic manifold T^*X . The corresponding twisted version goes as follows.

Definition 1.7.6. Given Atiyah algebra $\mathcal{A}$, let $\text{Sym}(\mathcal{A}/\mathcal{O})$ denotes the sheaffication of the quotient of $T_{\mathbb{C}}(\mathcal{A})$ by the relations:

$$D_1 \otimes D_2 - D_2 \otimes D_1 = 0,$$

$$f \otimes D_1 = f D_1,$$

and

$$1_{T_{\mathbb{C}}(\mathcal{A})} = 1_{\mathcal{A}},$$

for all $f \in \mathcal{O}, D_{1,2} \in \mathcal{A}$.

It follows that the sheaf of $\mathcal{O}_X$ -algebras $\text{Sym}(\mathcal{A}/\mathcal{O})$ is filtered, we have $\text{Sym}_{\leq 1}(\mathcal{A}/\mathcal{O})$. The natural map

$$\mathcal{A} \rightarrow \text{Sym}(\mathcal{A}/\mathcal{O}),$$

induces, by transferring the Lie bracket, a Poisson structure such that

$$\{-, -\}: \text{Sym}_{\leq k}(\mathcal{A}/\mathcal{O}) \otimes \text{Sym}_{\leq m}(\mathcal{A}/\mathcal{O}) \rightarrow \text{Sym}_{\leq k+m-1}(\mathcal{A}/\mathcal{O}).$$

Passing to the associated graded is equivalent to “killing” the twisting:

$$\bigoplus_k \text{Sym}_{\leq k}(\mathcal{A}/\mathcal{O}) / \text{Sym}_{< k}(\mathcal{A}/\mathcal{O}) \simeq \text{Sym}_{\mathcal{O}}(TX) = \mathcal{O}(T^*X).$$

Definition 1.7.7. The extension class of $\mathcal{O}_X$ -modules

$$0 \rightarrow \mathcal{O} \rightarrow \mathcal{A} \rightarrow TX \rightarrow 0,$$

denoted by $\text{at}(\mathcal{A}) \in \text{Ext}_X^1(TX, \mathcal{O}) = H^1(\Omega^1)$, is called *Atiyah’s class* of $\mathcal{A}$.

Recall that $\text{Diff}_{\leq 1}(L) \simeq \text{Hom}(J^{\leq 1}(L), L)$, where $J^{\leq k}(-)$ is the bundle of k -jets. It is convenient to consider the Atiyah’s class $\text{at}(\text{Diff}_{\leq 1}(L))$ in terms of the dual exact sequence:

$$0 \rightarrow \Omega^1 \rightarrow \text{Hom}(L, J^{\leq 1}(L)) \rightarrow \mathcal{O} \rightarrow 0, \quad (1.7.8)$$

treated as Ω^1 -torsor of connections on L , or equivalently splittings $\mathcal{O} \rightarrow \text{Hom}(L, J^{\leq 1}L)$. In particular the class $\text{at}(\text{Diff}_{\leq 1}(L))$ is the obstruction for an existence of a (holomorphic) connection on L .

Definition 1.7.9. The Ω^1 -torsor corresponding to the Atiyah’s class $\text{at}(\mathcal{A}) \in \text{Ext}_X^1(\mathcal{O}, \Omega^1)$ is called *twisted cotangent bundle* and is denoted by $T_{\mathcal{A}}^*X$.

Exercise Check that C^∞ -splitting of (1.7.8), is equivalent to $(1, 0)$ -connection

$$\nabla^{1,0}: L \rightarrow \Lambda^{1,0} \otimes L.$$

Deduce that $[\bar{\partial}, \nabla^{1,0}]$ represents the class

$$\text{at}(\text{Diff}_{\leq 1}(L)) \in \text{Ext}^1(\mathcal{O}, \Omega^1) \simeq H^{1,1}(X).$$

and that for Kähler X , the class $\text{at}(\text{Diff}_{\leq 1}(L))$ is equal to $-2\pi i \cdot c_1(L)$.

Proposition 1.7.10. *The projection $p: T_{\mathcal{A}}^*X \rightarrow X$ is a Lagrangian fibration with respect to a natural symplectic form. The map $\mathcal{A} \rightarrow p_*\mathcal{O}(T_{\mathcal{A}}^*X)$ induces an isomorphism of filtered Poisson algebras*

$$\mathrm{Sym}(\mathcal{A}/\mathcal{O}) \xrightarrow{\sim} p_*\mathcal{O}(T_{\mathcal{A}}^*X).$$

Proof. 1. The set of trivialization of Atiyah's algebra $\mathcal{A}$ over some open $U \subset X$, is a torsor over $\Omega_{cl}^1(X)$. Such a trivialization tautologically induces a splitting $TX \rightarrow \mathcal{A}$ of (1.7.1), this corresponds to the inclusion $\Omega_{cl}^1 \rightarrow \Omega^1$. On the other hand, the splitting $TX \rightarrow \mathcal{A}$ is equivalent to the identification $\phi: T_{\mathcal{A}}^*|_U \simeq T^*X|_U$ of T^*X -torsors over U . Denote by ω holomorphic 2-form on T^*X . Since ϕ is chosen up to a closed 1-form, symplectic form $\phi^*(\omega) \in \Omega^2(T_{\mathcal{A}}^*)$ doesn't depend on any choice.

2. Since $T_{\mathcal{A}}^*X$ is a torsor over vector bundle T^*X , the sheaf of algebras $p_*\mathcal{O}(T_{\mathcal{A}}^*X)$ is filtered by the degree of polynomial functions along the affine fibers. The natural map $\mathcal{A} \rightarrow p_*\mathcal{O}(T_{\mathcal{A}}^*X)$ extends to a map of algebras $\mathrm{Sym}(\mathcal{A}/\mathcal{O}) \rightarrow p_*\mathcal{O}(T_{\mathcal{A}}^*X)$. In the untwisted case, both parts are isomorphic to $\mathrm{Sym}_{\mathcal{O}} TX$. Since $\mathcal{A}$ is locally trivial the claim follows. $\square$

Proposition 1.7.11. *The automorphism group of trivial Atiyah's algebra $\mathrm{Diff}_{\leq 1}(\mathcal{O})$ is isomorphic to the sheaf of closed 1-forms Ω_{cl}^1 . The isomorphism classes of Atiyah's algebras are given by classes in $H^1(X; \Omega_{cl}^1)$.*

Proof. An automorphism of $\mathcal{O}_X \oplus TX$ has a form $(\mathrm{id}, \mathrm{id} + \eta)$, where $\eta: TX \rightarrow \mathcal{O}_X$ is a 1-form. The compatibility with the Lie bracket is equivalent to the condition

$$a.\eta(b) - b.\eta(a) - \eta([a, b]) = 0,$$

for any pair of vector fields a, b . In other words η is closed. Given arbitrary Atiyah's algebra $\mathcal{A}$ satisfying the definition 1.7.1, choose a Čech covering $\sqcup U_i \rightarrow X$ with trivialization of the restriction $\mathcal{D}_{\leq 1}|_{U_i}$. This induces 1-cocycle $\eta_i \in \Omega_{cl}^1(U_i)$. The standard argument shows that two central extensions are isomorphic $\iff$ the corresponding cocycles differ by a coboundary, corresponding to change of the trivialization on opens U_i . $\square$

Hence the isomorphism classes of Atiyah's algebras form an abelian group. One can check that in particular we have the following formula for the sum

$$[\mathrm{Diff}_{\leq 1}(L)] + [\mathrm{Diff}_{\leq 1}(M)] = [\mathrm{Diff}_{\leq 1}(L \otimes M)].$$

Similarly it allows to define $[\mathrm{Diff}_{\leq 1}(L^{\otimes s})]$ for arbitrary complex number $s \neq 0$ by the sequence:

$$0 \rightarrow \mathcal{O} \xrightarrow{i/s} \mathrm{Diff}_{\leq 1}(L) \rightarrow TX \rightarrow 0, \quad (1.7.12)$$

representing the class $s \cdot [\mathrm{Diff}_{\leq 1}(L)]$. If $s \in \mathbb{Z} \setminus \{0\}$, then one has the following isomorphism of short exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{O} & \xrightarrow{i/s} & \mathrm{Diff}_{\leq 1}(L) & \longrightarrow & TX \longrightarrow 0 \\ & & \downarrow & & \downarrow \phi & & \downarrow \\ 0 & \longrightarrow & \mathcal{O} & \xrightarrow{i} & \mathrm{Diff}_{\leq 1}(L^{\otimes s}) & \longrightarrow & TX \longrightarrow 0 \end{array} \quad (1.7.13)$$

where for any $D \in \text{Diff}_{\leq 1}(L)$, $\phi(D)$ applied to $e_1 \otimes \dots \otimes e_s \in L^{\otimes s}$ is equal to

$$\sum_{i \leq s} e_1 \otimes \dots \otimes D(e_i) \otimes \dots \otimes e_s.$$

In general, the abelian group structure on Atiyah's algebras can be described in terms of the short exact sequence, but we don't need this.

Note that the Lie algebra of $\mathbb{C}^\times$ -invariant vector fields on $\mathbb{C}^\times$ -bundle $L^\times$, viewed as a sheaf over X , fits into exact sequence

$$0 \rightarrow \mathcal{O}_X \xrightarrow{i} T(L^\times)^{\mathbb{C}^\times} \xrightarrow{p} TX \rightarrow 0,$$

where i sends 1 to the vertical Euler vector field $t\partial_t$ induced by the action of $\mathbb{C}^\times = \text{Spec } \mathbb{C}[t, t^{-1}]$ on $L^\times$.

Proposition 1.7.14. *There is natural isomorphism of Atiyah's algebras*

$$\text{Diff}_{\leq 1}(L) \simeq T(L^\times)^{\mathbb{C}^\times}.$$

The vertical map in (1.7.13) induces an isomorphism

$$\text{Diff}(L^{\otimes \lambda}) \simeq \text{Diff}(L^\times)^{\mathbb{C}^\times} / (t\partial_t - \lambda),$$

where $\text{Diff}(L^\times)^{\mathbb{C}^\times}$ is the ring of $\mathbb{C}^\times$ -invariant ordinary differential operators on $L^\times$.

Proof.

□

1.8 Modules over TDO's and projectively flat connections

Definition 1.8.1. A module M over sheaf of Lie algebras $\mathcal{A}$ is called *Atiyah's module*, if $\mathcal{O}_X \subset \mathcal{A}$ induces $\mathcal{O}_X$ -module structure on M .

In particular, by 1.7.12 Atiyah module over $\text{Diff}_{\leq 1}(L^{\otimes s})$ is a module M over Lie algebra $\text{Diff}_{\leq 1}(L)$ such that $\mathcal{O} \subset \text{Diff}_{\leq 1}(L)$ acts as $f \cdot m = sf \cdot m$.

There is equivalence of categories of Atiyah modules over, in general non-isomorphic Atiyah's algebras, $\text{Diff}_{\leq 1}(L^{\otimes s})$ and $\text{Diff}_{\leq 1}(L^{\otimes s+n})$ for any integer n . Indeed, if M is a module over Lie algebra $\text{Diff}_{\leq 1}(L)$, where $f \in \mathcal{O}$ acts by the multiplication by sf , then $M \otimes L^{\otimes n}$ is a module over Lie algebra $\text{Diff}_{\leq 1}(L)$ such that f acts by multiplication by $(s+n)f$. Hence the functor

$$M \rightsquigarrow M \otimes L^{\otimes n},$$

provides the required equivalence.

Proposition 1.8.2. *There is equivalence between Atiyah modules over $\mathcal{A}$ and modules over the associative algebra $\mathcal{U}(\mathcal{A}/\mathcal{O})$.*

Corollary 1.8.3. *There is equivalence between flat bundles over $L^\times$ with the monodromy along the fiber equal to $\exp(-2\pi i \cdot \lambda)$ and $\text{Diff}(L^\lambda)$ -modules which are coherent as $\mathcal{O}_X$ -sheaves.*

Proof. Assume M is a module over $\mathcal{U}(\mathcal{A}/\mathcal{O})$. By 1.7.14, M is a module over $p_* \text{Diff}(L^\times)^{\mathbb{C}^\times}$ such that $t\partial_t$ acts as $\lambda \cdot \text{id}$. Consider $\text{Diff}(L^\times)$ -module

$$\tilde{M} = \text{Diff}(L^\times) \otimes_{p_* \text{Diff}(L^\times)^{\mathbb{C}^\times}} M.$$

Locally, over some open U , one can assume that L is trivial. Chose coordinates x_i over U and t a coordinate on L . Then $p_* \text{Diff}(L^\times)^{\mathbb{C}^\times}$ is an algebra generated by $t\partial_t, \partial_{x_i}$ over $\mathcal{O}(U)$ (functions in x_i). It is easy to see that if M was free $\mathcal{O}_X$ -module, then $\tilde{M}$, as $\mathcal{O}(L^{\text{times}})$ -module is isomorphic to $\mathbb{O}(L^\times|_U) \otimes_{\mathcal{O}(U)} M$. In particular, if $\tilde{M}$ is $\text{Diff}(L^\times)$ -module, which is coherent $\mathcal{O}(L^\times)$ -module, hence by Katz theorem it is free $\mathcal{O}(L^\times)$ -module of finite rank. It follows also that M is free $\mathcal{O}_X$ -module of the same rank as $\tilde{M}$. Consider $\tilde{M}$ restricted to the fiber $F = \{x_i = a_i\} \simeq \mathbb{C}^\times$, it is a flat bundle such that $M \subset \tilde{M}$ is given by sections $s \in \tilde{M}|_F$ such that $t\partial_t.s = \lambda \cdot s$. Since $\tilde{M}|_F = \mathbb{C}[t] \otimes_{\mathbb{C}} M|_F$, the monodromy of the D-module $\tilde{M}|_F$ is equal to $\exp(-2\pi i \cdot \lambda)$.

We claim that the functor $M \rightsquigarrow \tilde{M}$ is equivalence of categories. Assume $\tilde{M}$ is a D-module over $L^\times$, such that fiber wise monodromy is equal to $\exp(-2\pi i \cdot \lambda)$. Define M as a subsheaf of sections s such that $t\partial_t.s = \lambda \cdot s$. It is easy to check that this construction provides the inverse functor. $\square$

The exact sequence of sheaves

$$1 \rightarrow \mathbb{C} \rightarrow \mathcal{O} \xrightarrow{d} \Omega_{cl}^1 \rightarrow 0,$$

implies the isomorphism

$$H^1(\Omega_{cl}^1) \xrightarrow{\sim} \ker(H^2(X; \mathbb{C}) \rightarrow H^2(X; \mathcal{O})).$$

If X is Kähler, then we obtain

$$H^1(\Omega_{cl}^1) \simeq H^0(X; \Omega^2) \oplus H^1(X; \Omega^1).$$

The projection $H^1(X; \Omega_{cl}^1) \rightarrow H^1(X; \Omega^1)$ is induced by the inclusion of sheaves: it maps $\mathcal{D}_{\leq 1}$ to the ext's class of the extension 1.7.1 of $\mathcal{O}_X$ -modules, called the Atiyah's class. The form $w \in H^0(X; \Omega^2)$ is automatically closed if X is Kähler, and corresponds to $\mathcal{O} \oplus TX$ with the bracket $[a, b]_w = [a, b] + w(a, b)$.

1.9 Moduli of bundles

See [AB83],[Hit87]. **TO BE ADDED**

2 Conformal Field Theory

On the general theory see [BPZ84],[MS90], [Seg88],[Mne25]. One should provide some necessary facts about the representation theory [FB04],[Kac90]. For the relation of conformal blocks to TQFT see [Wit84],[Kon88].

TO BE ADDED

3 Quantization of Chern-Simons

See [Wit89],[Eli+89], [Mne25], [BK+01],[Kon88].

3.1 Path integrals in a nutshell

1. Fix a finite dimensional vector space V over reals together with an Euclidian metric $(-, -)$. The metric defines a volume form such that the unit cube has unit volume. We denote the corresponding measure by $\mathcal{D}a$. Assume $D: V \rightarrow V$ is a positive self-adjoint operator: $(Da, b) = (a, Db)$ for all $a, b \in V$ and $(Da, a) > 0$ for all $a \neq 0$.

We list three important corollaries of the identity $\int_{\mathbb{R}} e^{-\frac{1}{2}t^2} dt = \sqrt{2\pi}$. We will refer to elements of V as (classical) *fields* and call the function $S(a) = \frac{1}{2}(a, Da)$ the *action functional*. Let $k > 0$ be some parameter. We start with the *partition function*:

$$Z := \int_V e^{-kS(a)} \mathcal{D}a = \frac{1}{\sqrt{\det(kD/2\pi)}}. \quad (3.1.1)$$

One can introduce an auxiliary *source field* $f \in V$:

$$Z[f] := \int_V e^{-kS(a) + (f, a)} \mathcal{D}a = e^{\frac{1}{2}(f, (kD)^{-1}f)} \cdot Z. \quad (3.1.2)$$

Partial derivatives of $Z[f]$ can be used to compute *correlator functions*. For example

$$\langle (f, a), (g, a) \rangle := \frac{1}{Z} \int_V e^{-kS(a)} (f, a)(g, a) \mathcal{D}a = \frac{1}{Z} \delta_f \delta_g Z[f]|_{f=0} = (f, (kD)^{-1}g), \quad (3.1.3)$$

is the formula for *2-point correlator function*. In general, if $F(a)$ is any polynomial function in a we can define $\langle F(a) \rangle$. It is equal to the expectation value of $F(a)$ with respect to the probability measure $\det(kD/2\pi) \cdot e^{-\frac{1}{2}(a, kDa)} \mathcal{D}a$. An n -point function is computed by means of *Wick formula*:

$$\langle (f_1, a) \dots (f_n, a) \rangle = \sum_{\Gamma_n} \prod_{e \in E(\Gamma_n)} (f_{e_0}, (kD)^{-1}f_{e_1}),$$

where Γ_n is the set of graphs on labeled vertices $1, \dots, n$ with the set of edges $e \in E(\Gamma_n)$ such that none of different edges has an intersections. Here e_0, e_1 denotes unordered pair of end points of the edge e . Clearly n -point function vanish for odd n . One should think of $f_i \in V$ as a vertex labeled by f_i with an attached half edge. Wick formula says that one has sum over all the ways to glue half edges of different f_i , the resulting graph has vertices labeled by f_i and edges decorated by self-adjoint $(kD)^{-1}$.

The 2-point function determines all correlators, the inverse D^{-1} is called *propagator* and in practice often appear as a Green function.

Notice that if we forget about integrals in the middle, all formulas on the right hand side, by analytic continuation, make sense as long kD is non degenerate. For any complex number $k \neq 0$ and non degenerate D we *define* the integrals in the middle to be consistent with the formulas on the right hand sides. The assumption that V is finite dimensional was necessary only to define $\det(kD)$. One can try to overcome even this restriction by using *zeta regularization*: if $T_s := \text{Tr}[(kD)^s]$ is well-defined for some open domain in $\mathbb{C}$, then one can set $\frac{d}{ds} T_s|_{s=0} = \det(kD)$.

In practice people take $k = \frac{i}{\hbar}$ and pretend that we are working in quantum field theory setting with Planck's parameter $\hbar$, or consider real $k > 0$ as the temperature in statistical mechanics.

So far we considered only the simplest case with the quadratic action $S(a)$. Usually the action includes, akin to the quadratic kinetic term, an *interactions* of fields. The general action can be written as $S_I(a) = S(a) + I(a)$, where $I(a) = \sum_i I_i(a)$ and $I_i \in \text{Sym}^{d(i)}(V^*)$ for some degree function $d(i) > 2$. Here we allow different terms of the same degree. If we consider the term I as

the *perturbation* of S , then we can relate theory with interaction to the correlators of quadratic theory:

$$\langle F(a) \rangle_I = \frac{1}{Z_I} \int e^{-kS(a) - kI(a)} F(a) \mathcal{D}a = \frac{1}{Z_I/Z} \sum_n \frac{(-k)^n}{n!} \langle I(a)^n F(a) \rangle,$$

and

$$\frac{Z_I}{Z} = \sum_n \frac{(-k)^n}{n!} \langle I(a)^n \rangle.$$

Notice that degree d part in $I(a)^n$ is a sum over monomials $\prod_{i \leq n} I_i(a)^{k_i}$ with $\sum k_i d(i) = d$ and $\sum k_i = n$. The value $k^n \langle I_{i_1}(a) \cdot I_{i_n}(a) \rangle$ is given by a sum over all graphs $\Gamma = \Gamma_{i_1, \dots, i_t}$ with n vertices labeled by the tensor $I_{i_t} \in \text{Sym}^{d(i_t)}(V^*)$, $t \leq n$ of valency $\deg I_{i_t}$. The number of edges of Γ is equal to $\frac{1}{2} \sum_t d(i_t)$. If we decorate each edge of $\Gamma = \Gamma_{i_1, \dots, i_t}$ with the propagator $(kD)^{-1}$ one can compute the contribution of the term corresponding to Γ , similarly to the Wick's formula: it is equal, up to a combinatorial coefficient, to $k^{n - \frac{1}{2} \sum_t d(i_t)} \gamma_{i_1, \dots, i_t}(D, I)$, where $\gamma(D)$ depend only on D and interactions I . Here we used the fact that each edge contributes to k^{-1} and each vertex I_{i_t} by $k^{d(i_t)}$. Note that $\chi(\Gamma) = n - \frac{1}{2} \sum_t d(i_t) < 0$ because $d(i_t) > 2$. The graphs are called Feynman's diagrams. They provide an expansion in powers of k^{-1} , such that the terms of order k^χ are given by graphs Γ with Euler characteristics equal to χ . Here Γ allow loops, but without vertices of valence 1, 2, thus $\chi < 0$.

The similar considerations provide an expression for $\langle F(a) \rangle_I$ with $F \in \text{Sym}^d(V)$, as a sum over Feynman diagrams with d exterior half edges, which should be couple with the star shaped vertex F with d half edges.

This is sketch of the perturbative analysis of QFT. One can justify the approach by using change of variables $a \rightsquigarrow \frac{a}{\sqrt{k}}$ and noting that it is equivalent to working with $k = 1$ and the interaction $I(a, k) = \sum k^{-d(i)/2} I_i(a)$. One can expect that for large enough k the contribution of the term $I(a, k)$ with arbitrary small coupling constants implies arbitrary small changes of correlators. But it is not the case. For instance the non perturbative partition function with the action $S(t) = \frac{1}{2}t^2 + k^{-3/2}t^3$ doesn't make any sense for any $k > 0$. Thus the power series described with Feynman technique k^{-1} are formal and they don't converge.

If $I(a)$ is written in terms of monomials (f, a) for various $f \in V$, then $\langle F(a) \rangle_I$ admits a nice combinatorial expansion in powers of k^{-1} , it is called Feynman's diagrams expansion. The main feature of this expansion is that its terms are parametrized by graphs with labeled vertices and their values are computed by means of propagator D^{-1} .

Let's go back to the quadratic theory and assume now that D is degenerate. The kernel $K = \ker D$ is the space of so called *zero modes*. Taking the orthogonal complement $K^\perp$ with respect to the inner product $(-, -)$, we obtain a decomposition $V = K \oplus K^\perp$. One can view the translation along K as a symmetry of the theory. All integrals of invariant expressions under the translation along K are equal to the integrals over $K^\perp$ times the volume of K .

There are two solutions to this problem. The first is to restrict our space of fields to $K^\perp$, where $D: K^\perp \rightarrow K^\perp$ is invertible. The second is to introduce a parameter m such that $D + m \cdot \text{id}$ is invertible, solve theory there to get an analytic expression in m and then pass to the limit $m \rightarrow 0$. In fact the both approaches are equivalent, this is related to the fact that

$$\sqrt{m/2\pi} e^{-\frac{1}{2}mt^2} \rightarrow \delta_0(t),$$

as $m \rightarrow 0$.

3.2 Example: abelian Chern-Simons over S^3 and linking number of knots

We will apply the previous discussion to the Chern-Simons theory with gauge group is $U(1)$ and $M = S^3$

A connection in $U(1)$ -bundle has a form $d + iA$, where $A \in \Lambda^1(M)$, and the action is given by $S'_{CS}(A) = \int_M A \wedge dA$. Since $S^3 = \mathbb{R}^3 \cup \{\infty\}$ one can consider any 1-form on S^3 as a 1-form on $\mathbb{R}^3$ with sufficiently fast decay on infinity.

Recall that a metric g on M defines the Hodge star-operator

$$*: \Lambda_M^i \rightarrow \Lambda_M^{\dim M - i},$$

defined by the equality $a \wedge *b = (a, b) \text{Vol}_g$, where $(-, -)$ is the induced metric on Λ^i and Vol_g is the associated volume form. We will consider $\mathbb{R}^3$ with coordinates $x^i, i = 1, 2, 3$ with the standard metric $g_{ij} = \delta_{ij}$. We will write $A = A_i dx^i \in \Lambda^1$, in particular $dA = d(A_i dx^i) = \partial_j A_i dx^j \wedge dx^i$. The Hodge operator is given by $*dx^i = \varepsilon_{ijk} dx^j \wedge dx^k$ and $*(dx^j \wedge dx^k) = \varepsilon_{jki} dx^i$, where ε_{ijk} is totally antisymmetric tensor determined by $\varepsilon_{123} = 1$.

Define a metric on the space of $U(1)$ -connections $\mathcal{C} := \mathcal{C}(M)$ by setting $(A, B) = \int_M A \wedge *B$ for any pair of 1-forms $A, B \in \Lambda^1(M)$. We can rewrite the action as $S'_{CS}(A) = (A, DA)$, where $D = *d$ is the curl operator on 1-forms. The partition function is equal to

$$Z = \int_{\mathcal{C}(M)} e^{ik'(A, DA)} \mathcal{D}A,$$

where $k' = \frac{k}{4\pi}$ for an integer k . The Wilson correlator on a collection of loops $\gamma_a: S^1 \rightarrow S^3$ corresponding to the characters $n_a: U(1) \rightarrow U(1)$ is equal to

$$\langle \prod_a \exp \left(in_a \int_{\gamma_a} A \right) \rangle = \frac{1}{Z} \int_{\mathcal{C}} \exp \left(ik'(A, DA) + \sum_a in_a \int_{\gamma_a} A \right) \mathcal{D}A,$$

where by definition $\int_{\gamma_a} A := \int_{S^1} \gamma_a^* A = \int_{S^1} A_i \dot{\gamma}_a^i dt$.

Note that $\ker D = \ker d$ is equal to the set of closed 1-forms on S^3 . Let $d^*: \Lambda^i \rightarrow \Lambda^{i-1}$ be the formal adjoint to the de Rham operator d , it is defined by the equality $(dA, B) = (A, d^*B)$. One has a formula $d^* = \pm * d *$. Using Hodge theory, it is one formally checks one has $\ker d = (\text{im } d^*)^\perp$ and dually $\ker d^* = (\text{im } d)^\perp$, where the orthogonal complements are with respect to the inner product $(-, -)$. Hence one can rewrite the orthogonal decomposition as

$$\Lambda^1(M) = \ker D \oplus \text{im}(d*),$$

so the restriction $D: \text{im}(d*) \rightarrow \text{im}(d*)$ is invertible.

Note that $\text{im } d \subset \Lambda_M^1$ corresponds to the infinitesimal gauge action: given $\delta g \in \Lambda_M^0$ one has $\delta A = d(\delta g)$. Since $H^1(S^3; \mathbb{R}) = 0$, zero modes of D coincide with infinitesimal gauge action. By the same argument $\text{im } D = \text{im}(d^*)$ is equal to $\ker d^* \subset \Lambda^1(M)$. The choice of orthogonal $\ker(d^*)$ to $\ker D \subset \Lambda^1(M)$ corresponds to well-known gauge fixing $d^* A = 0$ equivalent to $\sum \partial_{x^i} A_i = 0$.

The Green function of the curl operator on $\mathbb{R}^3$ is well-known. For $x, y \in \mathbb{R}^3$, set

$$G_{ij}(x, y) = \varepsilon_{ijk} \frac{1}{4\pi} \frac{x_k - y_k}{|x - y|^3}.$$

Given $f_j(x) dx^j \in \ker(d^*)$, let

$$g_i(x) = \int_{\mathbb{R}^3} G_{ij}(x, y) f_j(y) d^3 y.$$

Then a straightforward computation shows that

$$D(g_i(x)dx^i) = f_i(x)dx^i.$$

Let us redefine the path integral by integration only over the fields $A \in \ker(d*)$. As a corollary of (3.1.3) we obtain:

$$\langle A_i(x)A_j(y) \rangle = \frac{i}{2k} \varepsilon_{ijk} \frac{x_k - y_k}{|x - y|^3}.$$

Since the Wilson correlator is given by inserting the source term in the action, by (3.1.2) we obtain

$$\langle \prod_a \exp \left(i n_a \int_{\gamma_a} A \right) \rangle = \exp \left(\frac{i}{2k} \sum_{a,b} n_a n_b \int_{t_a, t_b \in S^1} \varepsilon_{ijk} \frac{\gamma_a^k(t_a) - \gamma_b^k(t_b)}{|\gamma_a(t_a) - \gamma_b(t_b)|^3} \dot{\gamma}_a^i \dot{\gamma}_b^j dt_a dt_b \right).$$

On the other hand one can check that

$$\Phi(\gamma_a, \gamma_b) = \frac{1}{4\pi} \int_{t_a, t_b} \varepsilon_{ijk} \frac{\gamma_a^k(t_a) - \gamma_b^k(t_b)}{|\gamma_a(t_a) - \gamma_b(t_b)|^3} \dot{\gamma}_a^i \dot{\gamma}_b^j dt_a dt_b = \int_{S^1 \times S^1} f^*(w_2),$$

where $f: S^1 \times S^1 \rightarrow S^2$ is the map $f(t_a, t_b) = \frac{\gamma_a(t_a) - \gamma_b(t_b)}{|\gamma_a(t_a) - \gamma_b(t_b)|}$ and w_2 is the volume form on S^2 rescaled such that $\int_{S^2} w_2 = 1$. Gauss integral formula says that $\Phi(\gamma_a, \gamma_b)$ is the linking number of knots γ_a and γ_b . Hence we obtain

$$\langle \prod_a \exp \left(i n_a \int_{\gamma_a} A \right) \rangle = \exp \left(\frac{2\pi i}{k} \sum_{a,b} \Phi(\gamma_a, \gamma_b) \right).$$

It follows that the Wilson correlators of abelian Chern-Simons on S^3 are equivalent to linking numbers modulo $k \in \mathbb{Z}$.

Strickly speaking the integrals $\Phi(\gamma_a, \gamma_b)$ might diverge. Though the Willson observables classically make sense for any maps $\gamma_a: S^1 \rightarrow M$, the quantum expectation values force us to assume that $\cup_a \gamma_a$ is a disjoint union of knots, i.e. a link. In this case one still has a short distance divergence when $a = b$.

In order to avoid the problem the natural solution is to work with framed links instead. A frame of a knot is the trivialization of its normal bundle. Equivalently it is an infinitesimal deformation of the knot with no fixed points. We will denote a framed knot by $(\gamma, \delta\gamma)$, so $\gamma + \delta\gamma$ is a knot infinitesimally close to γ such that the intersection number $\Phi(\gamma, \gamma + \delta\gamma)$ make sense. Given a framed knot $(\gamma, \delta\gamma)$ we define its self-intersection number by $\Phi(\gamma, \gamma + \delta\gamma)$.

3.3 Decorated cobordisms and Wilson observables

Let us show that the Wilson observables are sections of certain equivariant bundles, obtained via pullback from $\mathcal{C}(\partial M)$ similarly to 1.4.3. Moreover we would like a description which make sense for 3-manifolds with boundary.

1. We start by enlarging the category of surfaces and cobordisms. Now by a surface we understand a surface *decorated* with representations of $SU(N)$ at some number of marked points. A cobordism between from decorated surface Σ to Σ' , can be though as the trace of an evolution Σ . Thus it is a 3-manifold M with a disjoint union of circles in the interior and intervals with the boundary in $\partial M = \Sigma \sqcup \Sigma'$. Each circle and interval is labeled by a representation. In fact the

quantum Chern-Simons theory will make sense only if we put even more structure. The point $p \in \Sigma$ labeled by a representation $R \in \text{SU}(N)$, should be provided with a *framing*, i.e. non zero tangent vector at p . Similarly, the circles and intervals should be framed as well, see 3.2. Let us formalize the idea in the non framed case. The framed case is left to the reader.

Definition 3.3.1. Decorated surface $(\Sigma, (p_k, R_k), k \in K)$ is a closed oriented surface Σ with disjoint set of points $p_k \in \Sigma$ labeled by the representations $R_k \in \text{Rep}(\text{SU}(N))$ over the index set K .

The vector space

$$R(\Sigma, (p_k, R_k), k \in K) = \otimes_{k \in K} R_k, \quad (3.3.2)$$

can be thought as a vector bundle over configuration space of $|K|$ points over Σ . Moreover, if $g \in \text{Maps}(\Sigma, \text{SU}(N))$, then the evaluation $\text{ev}_{p_k}(g) \in \text{SU}(N)$ defines a representation

$$R(\Sigma, (p_k, R_k), k \in K) \in \text{Rep}(\text{Maps}(\Sigma, \text{SU}(N))). \quad (3.3.3)$$

If (Σ, p_k, R_k) is a decoration, then we define the inversion of the orientation by

$$-(\Sigma, p_k, R_k) = (-\Sigma, p_k, R_k^\vee),$$

where $-\Sigma$ denotes Σ with opposite orientation.

Definition 3.3.4. Decorated 3-manifold $(M, \partial M, (\gamma_i, R_i), i \in I \sqcup I^\circ)$, consists of 3-dimensional manifold $(M, \partial M)$, the index sets of *knots* I and the *intervals* I° , and the embedding

$$\bigsqcup_{i \in I} S^1 \sqcup \bigsqcup_{j \in I^\circ} [0, 1] \rightarrow M,$$

with the components given by the knots

$$\gamma_i: S^1 \rightarrow M \setminus \partial M, \quad i \in I,$$

and the intervals

$$\gamma_j: ([0, 1], \{0, 1\}) \rightarrow (M, \partial M), \quad j \in I^\circ.$$

Given M as above, the (possibly non connected) surface ∂M is naturally decorated by the *incoming* points $(\gamma_j(0), R_j^\vee)$ and the *outgoing* points $(\gamma_j(1), R_j)$, where j runs over I° . We parametrize the elements of this set by the formal disjoint union

$$K = I_{in}^\circ \sqcup I_{out}^\circ,$$

where $I^\circ = I_{in}^\circ = I_{out}^\circ$. The corresponding labeled points on the surface ∂M are written as $(p_k, R_k), k \in K$. The above definition says that the decorated surface

$$(\partial M, (p_k, R_k), k \in K),$$

is cobordant to $\emptyset$ via decorated cobordism M . Of course the notions of incoming and outgoing points on ∂M make sense only for the given cobordism M .

As usual, a cobordism can be viewed as a morphism in different ways. We will give the formal definition in the decorated setting for clarity.

Definition 3.3.5. Decorated cobordism

$$(\Sigma, (p_k, R_k), k \in K) \rightarrow (\Sigma', (p'_{k'}, R'_{k'}), k' \in K'),$$

is a decorated 3-manifold

$$(M, \partial M, (\gamma_i, R_i), i \in K \sqcup K' \sqcup I^\circ),$$

provided with an identification

$$-(\Sigma, (p_k, R_k)) \sqcup (\Sigma, (p'_{k'}, R'_{k'})) \xrightarrow{\sim} (\partial M, (\gamma_k(0), R_k^\vee), (\gamma_{k'}(1), R_{k'})), \quad k \in K, k' \in K'$$

2. Here we generalize the classical Wilson observables to the non-abelian case with boundary. Suppose

$$(M, \partial M, (\gamma_i, R_i), i \in I \sqcup I^\circ) \quad (3.3.6)$$

is a decorated 3-manifold with a connection $\nabla = d + A \in \mathcal{C}(M)$, where $\mathcal{C}(M) = \mathcal{C}(M, \text{SU}(N))$ is the space of $\text{SU}(N)$ -connections on the trivial bundle. Denote the induced decorated boundary surface $(\partial M, (p_k, R_k), k \in K = I_{in}^\circ \sqcup I_{out}^\circ)$, simply by ∂M .

For a map $\gamma: [0, 1] \rightarrow M$, the parallel transport with respect to $d + A$ along $[0, 1]$, is $\text{Pexp}(-A(\dot{\gamma})) \in \text{SU}(N)$. Given a representation $R \in \text{Rep}(\text{SU}(N))$, we denote the corresponding operator acting on R by

$$[\text{Pexp}(-A(\dot{\gamma}))|R] \in R^\vee \otimes R = \text{Hom}_{\mathbb{C}}(R, R).$$

We define the Wilson observable of (3.3.6) at $\nabla = d + A \in \mathcal{C}(M)$ by

$$W(M, \partial M, (\gamma_i, R_i))_\nabla = \prod_{i \in I} \text{tr}[\text{Pexp}(-A(\dot{\gamma}_i))|R_i] \cdot \prod_{j \in I^\circ} [\text{Pexp}(-A(\dot{\gamma}_j))|R_j], \quad (3.3.7)$$

viewed as an element of

$$R(\partial M) = \bigotimes_{k \in K} R_k = \bigotimes_{j \in I^\circ} (R_j^\vee \otimes R_j).$$

3. Lets turn to the equivariance of Wilson observables with respect to gauge transformation $g: M \rightarrow \text{SU}(N)$. By (3.3.3), $R(\partial M)$ admits a natural action by $\text{Maps}(\partial M, \text{SU}(N))$ provided by the evaluation of g at the labeled points $p_k \in \partial M$ for $k \in K = I_{in}^\circ \sqcup I_{out}^\circ$.

Proposition 3.3.8. *One has*

$$W(M, \partial M, (\gamma_i, R_i))_{g.A} = g.W(M, \partial M, (\gamma_i, R_i))_A \in R(\partial M),$$

for all $g \in \text{Maps}(M, \text{SU}(N))$.

Proof. It is enough to treat components of the link in M , or equivalently the terms in (3.3.7), separately. For a map $\gamma: [0, 1] \rightarrow M$, consider $B = -\gamma^*(A) = -A(\dot{\gamma})$ as a map $B: [0, 1] \rightarrow \mathfrak{su}(N)$ with values in matrix Lie algebra. The parallel transport $h(t) = \text{Pexp}(-A(\dot{\gamma}))$, viewed as a map $[0, 1] \rightarrow \text{SU}(N)$, is equal to $h(t)h(0)^{-1}$, where h is any solution to the differential equation

$$\dot{h} = Bh.$$

If $g: [0, 1] \rightarrow \text{SU}(N)$ is any map, then $h' = gh$ satisfies the equation

$$\dot{h}' = (\dot{g}g^{-1} + gBg^{-1})h',$$

and hence

$$h'(t)h'(0)^{-1} = g(t) \cdot (h(t)h(0)^{-1}) \cdot g^{-1}(t).$$

Thus the terms in (3.3.7) corresponding to the knots are invariant, because the trace is conjugate invariant, while the terms corresponding to the intervals, transform according to the representation $R^\vee \otimes R = \text{Hom}_\mathbb{C}(R, R)$. This proves the main assertion. $\square$

For any decorated surface $(\Sigma, (p_k, R_k), k \in K)$ by 1.5.2, the representation

$$R(\Sigma, (p_k, R_k), k \in K) = \bigotimes_k R_k \in \text{Rep}(\text{Maps}(\Sigma, \text{SU}(N))),$$

defines a unitary bundle

$$\mathcal{W}(\Sigma, (p_k, R_k), k \in K),$$

over the quotient $\mathcal{C}(\Sigma)/\text{Maps}(\Sigma, \text{SU}(N))$. Recall that

$$p: \mathcal{C}(M)/\text{Maps}(M, \text{SU}(N)) \rightarrow \mathcal{C}(\partial M)/\text{Maps}(\partial M, \text{SU}(N)),$$

is the restriction map and $\mathcal{L}$ denotes the natural $\text{U}(1)$ -bundle over $\mathcal{C}(\partial M)$ (1.4.4).

Let $\Sigma = \partial M$ and $K = I_{in}^\circ \sqcup I_{out}^\circ$.

Corollary 3.3.9. *The function*

$$e^{ikS_{CS}} \cdot W(M, \partial M, (\gamma_i, R_i), i \in I \sqcup I^\circ),$$

naturally descends to a section of unitary bundle

$$p^*(\mathcal{L}^{\otimes k} \otimes \mathcal{W}(\partial M, (p_k, R_k), k \in K)).$$

3.4 Canonical quantization

See [Eli+89]. **TO BE ADDED**

3.5 Jones polynomial

It is well-covered in [Wit89]. Possibly it make sense to clarify the framing correction in the Witten's article. **TO BE ADDED**

3.6 Modular functor and TQFT

It is the content of unpublished work [Kon88]. **TO BE ADDED**

3.7 S-matrix and Verlinde's coupling constants in Chern-Simons

Assume $Z: \text{Cob}_{2+1} \rightarrow \mathbf{Vect}$ is a decorated TQFT of Chern-Simons type. Consider a closed 3-dimensional manifold M . Suppose γ_i is a framed knot in M colored by some i . Here M can be decorated as well, we suppose that γ_i is separated from the intrinsic decoration. Let $D^2 \times S^1 \rightarrow M$ be the tube around γ_i . Let $M^\circ = M \setminus D^2 \times S^1$ and take $K \in \text{MCG}(S^1 \times S^1)$.

Recall that *the surgery along γ_i with respect to K* is a closed decorated 3-manifold

$$M[\gamma_i, K] := M^\circ \sqcup_{K: S^1 \times S^1 \xrightarrow{\sim} \partial(D^2 \times S^1)} D^2 \times S^1.$$

Here the result of the surgery has the same number of link components. The case $i = 0$ corresponds to surgery which doesn't affect the decoration in M . For instance $M[\gamma_i, \text{id}] = M$.

2. Let $v_i \in Z(S^1 \times S^1)$ denotes the vector $Z(D^2 \times S^1, \gamma_i)$. We have

$$Z(M[\gamma_i, K]) = \langle Z(M^\circ) | K Z(D^2 \times S^1, \gamma_i) \rangle = \langle Z(M^\circ) | K_i^j v_j \rangle = K_i^j \langle Z(M^\circ) | Z(D^2 \times S^1, \gamma_j) \rangle,$$

hence

$$Z(M[\gamma_i, K]) = K_i^j Z(M, \gamma_j). \quad (3.7.1)$$

In what follows we will be interested in the action of $S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in \text{MCG}(S^1 \times S^1)$.

The entries S_i^j are called the *S-matrix*. The map $C = S^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ of $S^1 \times S^1$ changes orientation of the time and space directions, it and it extends to $D^2 \times S^1$ by a π -rotation of the solid torus embedded in $\mathbb{R}^3$. Hence $Cv_i = v_{\bar{i}}$, because the change of the knot orientation is equivalent to conjugation of its label, which is evident from Wilson loops interpretation. The following *unitarity* property follows

$$\sum_j S_{i\bar{j}} S_{jk} = \delta_{ik}. \quad (3.7.2)$$

Recall that S^3 is a union of two solid tori $D^2 \times S^1$ glued by $S: S^1 \times S^1 \rightarrow S^1 \times S^1$. Denote by $(S^2 \times S^1, ijk)$ the embedding of parallel time-like knots in $S^2 \times S^1$. Also let (S^3, ij) be the Hopf link of two circles labeled by i, j . Similarly (S^3, i) will be unknotted circle labeled by i . The *Verlinde's coupling constants* are given by the symmetric tensor

$$N_{ijk} = Z(S^2 \times S^1, ijk) = \dim Z(S^2, ijk).$$

We have

$$(S^2 \times S^1)[\gamma_i, S] = (S^3, i),$$

and more generally

$$(S^2 \times S^1, \gamma_i)[\gamma_j, S] = (S^3, ij).$$

Recall that $Z(S^2 \times S^1, ij) = \dim Z(S^2, ij) = \delta_{i\bar{j}}$.

Let $(S^3, L(i, j, k)) = (S^2 \times S^1, jk)[i, S]$, it is a link formed by two small unknotted circles labeled by j, k around unknotted i .

$$Z(S^3, L(i, j, k)) = S_i^t Z(S^2 \times S^1, tjk) = S_i^t N_{tjk}. \quad (3.7.3)$$

Using the fact that $Z(S^2, i\bar{i})$ is 1-dimensional, by using a cut of S^3 along S^2 separating j and k , it is easy to show the following multiplicativity property

$$Z(S^3, L(ijk)) \cdot Z(S^3, i) = Z(S^3, ij) \cdot Z(S^3, ik). \quad (3.7.4)$$

Since $Z(S^2 \times S^1, ij) = \dim Z(S^2, ij) = \delta_{i\bar{j}}$, we have

$$Z(S^3, ij) = S_i^t Z(S^2 \times S^1, tj) = S_i^{\bar{j}} = S_{ij},$$

where we used the standard index notation with respect to the form $\delta_{i\bar{j}}$. By topological invariance, we obtain $S_{ij} = S_{ji}$. Similarly, since

$$Z(S^3, ij) = Z(-S^3, \bar{i}j) = Z(-S^3, i\bar{j}),$$

we have $S_{ij} = S_{i\bar{j}}$, i.e. S_j^i is symmetric matrix. Finally

$$Z(S^3, i) = S_i^t Z(S^2 \times S^1, t) = S_i^0 = S_{i0}.$$

Substitution of (3.7.4) to (3.7.3) then gives

$$\sum_t (S_i^t N_{tjk}) = \frac{S_{ij} S_{ik}}{S_{i0}},$$

multiplying by the symmetric inverse matrix $(S^{-1})_m^i$, gives the *Verlinde's formula*

$$N_{jk}^m = \sum_i \frac{S_j^i S_k^i (S^{-1})_i^m}{S_0^i}. \quad (3.7.5)$$

By using unitarity $(S^{-1})_j^m = S_j^m$ the above formula takes the more convenient form

$$N_{jk}^m = \sum_i \frac{S_{ij} S_{ik} S_i^m}{S_{0i}}. \quad (3.7.6)$$

Lemma 3.7.7. *The vectors $p_a = S_{a0} S_a^i v_i$ are canonical projectors: $p_a p_b = \delta_{ab} p_a$ such that $p^a = \frac{1}{S_{a0}^2} p_a$.*

3.8 Demystification of Verlinde's formula

Assume V is an commutative semisimple algebra of dimension $k+1$, with a basis v_i , where i runs over some set containing distingushed element 0. Suppose $v_0 = 1$ is the unit and the product is given by $v_i \cdot v_j = N_{ij}^k v_k$. Denote the multiplication by v_i by $N_i: V \rightarrow V$. By our assumption the operators N_i admit simultaneous diagonalization, i.e. there is $S: V \rightarrow V$ such that

$$S N_i S^{-1}(v_j) = \lambda_{i,j} v_j,$$

for some non-zero $\lambda_{i,j}$.

Remark 3.8.1. The above representation is unique up to an action by permutations and diagonal matrices: $\Sigma_{k+1} \ltimes (\mathbb{C}^\times)^{k+1}$.

The diagonalizability of N_i via S is equivalent to the condition

$$S^{-1}(v_i) \cdot S^{-1}(v_j) = \delta_{ij} \cdot \alpha_i \cdot S^{-1}(v_i),$$

for some scalars α_i . Equivalently $p_{\bar{a}} = 1/\alpha_a \cdot S^{-1}(v_a)$ are canonical projectors in V . Denoting their duals by $p^{\bar{a}} \in V^\vee$, we identify

$$p^{\bar{a}} = \varepsilon(p_{\bar{a}}) p_{\bar{a}} = (S^{-1})_a^0 / \alpha_a \in V, \quad (3.8.2)$$

in terms of the bilinear form

$$B := \varepsilon \circ m = \sum_a (S^{-1})_a^0 / \alpha_a \cdot p^{\bar{a}} \otimes p^{\bar{a}},$$

which is automatically non-degenerate.

Lemma 3.8.3. *Assume V as above, let $\varepsilon: V \rightarrow \mathbb{C}$ be the map $\varepsilon(v_i) = \delta_{i0}$. We have*

1. $\lambda_{a,j} = S_a^j / S_0^j$.
2. $N_{ij}^k = \sum_t S_i^t S_j^t (S^{-1})_t^k / S_0^t$. (3.7.5)
3. $\alpha_a = 1/S_0^a$, thus $p_{\bar{a}} = S_0^a \cdot S^{-1}(v_a)$ are canonical projectors and $\varepsilon(p_{\bar{a}}) = S_0^a \cdot (S^{-1})_a^0$.

Proof. 1.

$$SN_a S^{-1}(v_j) = \lambda_{a,j} v_j \xrightarrow{S_k^j} SN_a(v_k) = S(N_{ak}^t v_t) = S_k^j \cdot \lambda_{a,j} v_j,$$

thus

$$N_{ak}^t S_t^j = \lambda_{a,j} S_k^j \xrightarrow{(S^{-1})_j^m} N_{ak}^m = \sum_j (S^{-1})_j^m S_k^j \lambda_{a,j},$$

which for $k = 0$ gives

$$\delta_a^m = \sum_j S_j^m S_0^j \lambda_{a,j} \xrightarrow{S_m^b} S_a^b = S_0^b \cdot \lambda_{a,b}.$$

2. By the above

$$N_{ak}^m = \sum_j \frac{(S^{-1})_j^m S_k^j S_a^j}{S_0^j}.$$

3. The equality $SN_a S^{-1}(v_b) = \lambda_{a,b} v_b$ is equivalent to

$$v_a \cdot S^{-1}(v_b) = \lambda_{a,b} S^{-1}(v_b),$$

applying $\sum_a (S^{-1})_c^a$ and substitution $\lambda_{a,b} = S_a^b / S_0^b$, we have

$$S^{-1}(v_c) \cdot S^{-1}(v_b) = \sum_a \frac{S_a^b}{S_0^b} (S^{-1})_c^a S^{-1}(v_b) = \frac{\delta_c^b}{S_0^b} S^{-1}(v_b).$$

Hence $\alpha_b = 1/S_0^b$. □

3.9 Verlinde's algebra in unitary TQFT

1.

Definition 3.9.1. Frobenius algebra is a commutative algebra V over $\mathbb{C}$ with the augmentation $\varepsilon: V \rightarrow \mathbb{C}$ such that $B(a, b) = \varepsilon(ab)$ is a non-degenerate form.

The following observation gives a practical way to work with Frobenius algebras.

Theorem 3.9.2. If $Z: \text{Cob}_{1+1} \rightarrow \mathbf{Vect}$ is a TQFT, then $Z(S^1)$ is a Frobenius algebra. Conversely, any Frobenius algebra gives a $1+1$ -dimensional TQFT.

Proof. The algebra structure on $Z(S^1)$ is given by looking at the pair-of-pants $D^2 \setminus D^2 \sqcup D^2$. For the details see [BK+01, p. 4.3.1]. □

Fix $(2+1)$ -dimensional TQFT $Z: \text{Cob}_{2+1} \rightarrow \mathbf{Vect}$. Denote by $Z': \text{Cob}_{1+1} \rightarrow \mathbf{Vect}$ the $(1+1)$ -dimensional TQFT given by $Z'(-) = Z(- \times S^1)$. By the previous $Z'(S^1) = Z(S^1 \times S^1)$ is a Frobenius algebra.

The multiplication is induced by looking at the pair-of-pants times S^1 . With respect to the filling $D^2 \times S^1$ of $S^1 \times S^1$, we refer to the standard loops $a = S^1 \times \text{pt}$, $b = \text{pt} \times S^1$ as the space-like and the time-like, respectively.

Recall that the action of the mapping class group on $H_1(S^1 \times S^1; \mathbb{Z}) = \mathbb{Z}\langle a, b \rangle$ provides an isomorphism

$$\pi_0 \text{Aut}_+(S^1 \times S^1) \simeq \text{SL}(2, \mathbb{Z}) = \text{SL}(\mathbb{Z}\langle a, b \rangle).$$

Thus $\text{SL}(2, \mathbb{Z})$ acts on the vector space $Z(S^1 \times S^1)$. Recall that the discrete group $\text{SL}(2, \mathbb{Z})$ is generated by the matrices $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ (in the basis given by a, b), subject to the relations: $S^2 = (ST)^3 = -\text{id}$. We will set $C = -\text{id}$. We refer to the induced action by T, S, C on $Z(S^1 \times S^1)$ as the *Dehn's twist*, *S-matrix* and *conjugation*, respectively. Note that since the action by C on $S^1 \times S^1$ extends to the filling $D^2 \times S^1$ (by rotation of the solid torus in $\mathbb{R}^3$ by π), the induced action on $Z(S^1 \times S^1)$ preserves the Frobenius algebra structure.

2. To motivate the following recall that in framed (i.e. decorated) Chern-Simons theory the space $Z(S^1 \times S^1)$ is spanned by v_i , where i runs over a set of colors I - certain simple representations of $\text{SU}(N)$.² Namely, the Wilson time-like loop with the standard framing determines $v_i \in Z(\partial(D^2 \times S^1))$. We have $Cv_i = v_{\bar{i}}$, where $\bar{i}$ is the dual representation. It is convenient to denote by $0 \in I$ the vacuum, i.e. the trivial representation.

Moreover we have $Tv_i = \exp(2\pi i h_i) v_i$ for some numbers h_i , where $h_0 = 0$. We call v_i the *primary field* with the *conformal weight* h_i . Note that $\varepsilon(v_i v_j) = \dim Z(S^2, i, j) = \delta_{i\bar{j}}$.

Definition 3.9.3. *Verlinde's algebra* is a Frobenius algebra provided with the basis $v_i \in V, i \in I$ parametrized by the set I such that $\varepsilon(v_i v_j) = \delta_{i\bar{j}}$. Moreover $\text{SL}(2, \mathbb{Z})$ acts on V in a compatible way as above.

Assume V is a Verlinde algebra. In the basis of primary fields $v_i \in V$ we the multiplication and S -matrix take the form $v_i \cdot v_j = N_{ij}^k v_k$ and $Sv_i = S_i^j v_j$. The bilinear form $(v_i, v_j) = \varepsilon(v_i v_j) = \delta_{i\bar{j}}$ allows to lift the indices in the standard manner: $S_j^i = S_{\bar{i}\bar{j}}$ and $N_{ij}^{\bar{k}} = N_{ijk}$. We denote by $N_i: V \rightarrow V$ the multiplication operator by v_i .

Record the following lemma.

Lemma 3.9.4. 1. The tensors S_{ij} and N_{ijk} are symmetric:

$$S_{ij} = S_{ji}, \quad N_{ijk} = N_{jki}, \quad N_{ijk} = N_{jik}.$$

2. $(Sv_i, v_j) = S_{ij}$ and $(N_i v_j, v_k) = N_{ijk}$.

3. The S -matrix and N_i are self-adjoint: $(Sv_i, v_j) = (v_i, Sv_j) = S_{ij}$.

3. Assume $Z: \text{Cob}_{2+1} \rightarrow \mathbf{Vect}$ is a TQFT with the Verlinde algebra $V = Z(S^1 \times S^1)$. Let $Z'(-) = Z(- \times S^1)$ denotes the corresponding $(1+1)$ -TQFT. Consider the multiplication $m: V \otimes V \rightarrow V$ and the comultiplication $\Delta: V \rightarrow V \otimes V$. Denote the bilinear form $\varepsilon \circ m$ by B . Notice that if V is semisimple, then V is abstractly isomorphic to the algebra $\oplus_{a \in I'} \mathbb{C}$, where I' is a set of the same cardinality as I . Hence the canonical projectors p_a form a natural basis in V . Then

$$B(-, -) = \sum_a p_a \otimes p^a = \sum_a \varepsilon(p_a) p^a \otimes p^a \in \text{Sym}^2(V^\vee).$$

We are interested in the dimension of conformal blocks on genus g surface Σ_g .

²or more generally simple objects of the corresponding modular fusion category.

Proposition 3.9.5. *We have*

$$\dim Z(\Sigma_g) = \text{Tr}[M^{g-1}|V],$$

where $M = m \circ \Delta(1) \in V$. If $w_a \in V$ is a basis of V and $w^a \in V$ its dual with respect to B , then $M = \sum_a w_a \cdot w^a$.

Proof. Let P denotes a torus with two discs cut. The operator $Z'(P)$ is equal to $m \circ \Delta$. Arrange $g - 1$ -tori along an imaginary circle and take the connected sum of neighbors, we will get Σ_g . Hence $Z'(\Sigma_g) = \text{Tr}[Z'(P)^{g-1}|V]$. It is easy to see that $Z'(P)$ commutes with the multiplication, hence $Z'(P)$ is the multiplication by $M = m \circ \Delta(1)$. By definition

$$B(\Delta(1), w_a \otimes w_b) = B(1, w_a w_b) = B(w_a, w_b),$$

hence $\Delta(1) = \sum_a w_a \otimes w^a$. It follows that $M = \sum_a w_a \cdot w^a$ □

As a corollary we have $M = \sum_{i \in I} v_i v_{\bar{i}}$, thus

$$\dim Z(\Sigma_g) = \text{Tr}[(\sum_i N_i N_{\bar{i}})^{g-1}|V].$$

In other words the structure constants N_{ijk} determine the dimension of conformal blocks.

Similarly, if V is semisimple and p_a are canonical projectors, then

$$M = \sum_a p_a \cdot p^a = \sum_a 1/\varepsilon(p_a) \cdot p_a$$

and

$$\dim Z(\Sigma_g) = \sum_a \varepsilon(p_a)^{-(g-1)}. \quad (3.9.6)$$

Using $\varepsilon(p_{\bar{a}}) = S_0^a (S^{-1})_a^0$ and the unitarity $(S^{-1})_a^b = S_{\bar{a}}^b = S_{b\bar{a}}$ we obtain

$$\varepsilon(p_{\bar{a}}) = S_{0\bar{a}} S_{0a}. \quad (3.9.7)$$

Furthermore, $\sum_a S_a^t S_{\bar{a}}^t = 1$ and $S_a^t = \lambda_{a,t} S_0^t$ implies $\sum_a \lambda_{a,t} \lambda_{\bar{a},t} (S_0^t)^2 = 1$, hence

$$(S_0^t)^2 = (\sum_a \lambda_{a,t} \lambda_{\bar{a},t})^{-2}. \quad (3.9.8)$$

3.10 Dimension of conformal blocks and S-matrix

Assume $Z: \text{Cob}_{2+1} \rightarrow \mathbf{Vect}$ is a TQFT of Chern-Simons type. The Verlinde's formula (3.7.5) says that S-matrix diagonalizes the multiplication. The S-matrix satisfies $S^2 = C$, hence $(S^{-1})_b^a = S_{ab}$, so $\varepsilon(p_a) = (S_{0a})^2$. The the formal lemma 3.8.3 and 3.9.5 gives an expression for the dimension of conformal blocks $\dim Z(\Sigma_g)$ in terms of S-matrix.

Corollary 3.10.1. *We have $\dim Z(\Sigma_g) = \sum_a (S_{0a})^{2-2g}$.*

Verlinde's formula 3.7.5 states that S-matrix recovers the multiplication structure constants N_{ij}^k . This observation works in the opposite direction as well. Suppose N_{ij}^k are known. Then the eigenvalues $\gamma_{i,j}$ of the operators N_i are known as well. Since $\gamma_{i,j} = S_i^j / S_0^j$, it is enough to recover S_{0j} . By (3.9.8) we have

$$(S_0^j)^2 = (\sum_a \lambda_{a,j} \lambda_{\bar{a},j})^{-2},$$

this determines S_0^j up to a sign ± 1 . This is already enough to evaluate $\dim Z(\Sigma_g)$. We can say more and resolve the indeterminacy in the sign by noting that in Chern-Simons theory one always has $S_{0j} > 0$. Indeed $S_{0j} = Z(S^3, j)$ is the invariant of an unknot in S^3 labeled by j . By unitarity $S_{0j} = \langle Z(D^3, j) | Z(D^3, j) \rangle$, where (D^3, j) is the arc in D^3 labeled by j and $Z(D^3, j) \in Z(S^2, j\bar{j})$ is a generator of a 1-dimensional vector space. Since the theory is physical, it is plausible to expect that the quantum states are hilbert spaces, hence $S_{0j} > 0$ because $S_{0j} = 0$ for some j would force S to be degenerate.

2. We will illustrate this approach in the case of $SU(2)$ at the level k . The relation of Chern-Simons TQFT to Wess-Zumino-Witten CFT implies that the Verlinde algebra $Z(S^1 \times S^1)$ is isomorphic to K_0 of the category of integrable representations of the current algebra $\widehat{sl}_2^k$ (a central extension of $sl_2[t, t^{-1}]$). The basis of primary fields v_i is given by highest-weight irreducible quotients L_i of Verma modules, where $i = 0, \dots, k$. The only relevant fact for the moment is that the product $v_i \cdot v_j$ corresponds to the *fusion tensor product* $L_i \otimes_f L_j$ given by the following formula

$$L_i \otimes_f L_j = \bigoplus_{k=|i-j|}^{\min(i+j, 2k-i-j)} L_k,$$

where the summation is over k such that $k - i - j$ is even. In particular we obtain (here $k = 4$):

$$N_1 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}.$$

If $x_i, 0 \leq i \leq k$ are components of an eigenvector of N_1 , one can assume that $x_0 = 1$. Let $f(t) = x_0 + x_1 t + x_2 t^2 + \dots$, then $N_1 \cdot x = \lambda x$ is equivalent to $(f(t) - 1)/t + t f(t) = \lambda f(t)$. Thus

$$f(t) = \frac{1}{1 - \lambda t + t^2} = \frac{1}{\lambda_1 - \lambda_2} \left(\frac{1/\lambda_2}{1 - t/\lambda_2} - \frac{1/\lambda_1}{1 - t/\lambda_1} \right),$$

where λ_1, λ_2 are roots of the denominator. Thus

$$x_i = \frac{\lambda_1^{i+1} - \lambda_1^{-(i+1)}}{\lambda_1 - \lambda_1^{-1}}.$$

The boundary condition $x_{k+1} = 0$ translates to $\lambda_1^{2(k+2)} = 1$, so

$$\lambda_1 = q_s := \exp\left(\frac{\pi i(s+1)}{k+2}\right),$$

where $s = 0, \dots, k$ (because $\lambda_1 \neq \lambda_2 = \lambda_1^{-1}$). This is the formula for the eigenvalues of N_1 , denote λ_1 by q_s .

It is not hard to see that the corresponding eigenvalues of N_i satisfies the same relations as x_i , so

$$\lambda_{a,s} = \frac{q_s^{a+1} - q_s^{-(a+1)}}{q_s - q_s^{-1}} = \frac{\sin\left(\frac{\pi(s+1)(a+1)}{k+2}\right)}{\sin\left(\frac{\pi(s+1)}{k+2}\right)}.$$

By (3.9.8) $(S_0^s)^{-2} = \sum_a \lambda_{a,s}^2$, and as one can check the later is equal to

$$\frac{-2(k+1)}{(q_s - q_s^{-1})^2} = \frac{(k+1)/2}{\sin\left(\frac{\pi(s+1)}{k+2}\right)^2},$$

thus, taking into account $S_0^s > 0$, we get $S_0^s = \sqrt{\frac{2}{k+2}} \sin(\frac{\pi(s+1)}{k+2})$, and thus:

$$S_a^s = \lambda_{a,s} S_0^s = \sqrt{\frac{2}{k+2}} \sin(\frac{\pi(s+1)(a+1)}{k+2}).$$

Finally we obtain

$$\dim Z(\Sigma_g) = \sum_a (S_0^a)^{2-2g} = \sum_{a=0}^k \left(\frac{k+2}{2}\right)^{g-1} \sin\left(\frac{\pi(a+1)}{k+2}\right)^{2-2g}.$$

Let us note that if Σ_g is a complex curve, then for $SU(N)$ -Chern-Simons we have

$$\dim Z(\Sigma_g) = \dim H^0(\text{Bun}(\Sigma_g, \text{SL}(N, \mathbb{C})), \mathcal{L}^{\otimes k}),$$

where $\mathcal{L}$ is the Quillen determinant line bundle on the stack of rank N bundles with trivialized determinant $\text{Bun}(\Sigma_g, \text{SL}(N, \mathbb{C}))$. Recall that by Atiyah-Bott[AB83] and Narasimhan-Seshadri [NS65], the corresponding good moduli space (of semistable points!) can be identified with the quotient of flat bundles $\mathcal{C}(\text{SU}(N))^{flat} / \text{Maps}(\Sigma_g, \text{SU}(N))$ provided with the complex structure induced by that on Σ_g . Under this identification the Quillen line bundle corresponds to the restriction to the flat loci of the equivariant bundle described in the first section.

4 Abelian varieties

See [Mum70].

4.1 Line bundles on abelian varieties

Let $V \simeq \mathbb{C}^n$ be a vector space with a lattice $\Gamma \subset V$ of rank $2n$. The quotient $A = V/\Gamma$ is a compact complex torus. The group of holomorphic line bundle over A is denoted by $\text{Pic}(A)$.

1. By the definition of line bundle we have $\text{Pic}(A) = H^1(A; \mathcal{O}^*)$. The exponential exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O} \xrightarrow{e^{2\pi i -}} \mathcal{O}^* \rightarrow 1,$$

induces a long exact sequence of cohomology:

$$0 \rightarrow H^1(A; \mathbb{Z}) \rightarrow H^1(A; \mathcal{O}) \rightarrow H^1(A; \mathcal{O}^*) \xrightarrow{c_1} H^2(A; \mathbb{Z}) \rightarrow H^2(A; \mathcal{O}) \quad (4.1.1)$$

By the definition $\text{Pic}^0(A) = \ker c_1$. Recall that

$$H^2(A; \mathbb{Z}) = \text{Hom}_{\mathbb{Z}}(\Lambda^2 \Gamma, \mathbb{Z}),$$

so any $w \in H^2(A; \mathbb{Z})$ can be extended to a 2-form on $\Gamma \otimes \mathbb{R} = V$. We call w to be of type $(1, 1)$ if $w(ix, iy) = w(x, y)$ for all $x, y \in V$. Any $(1, 1)$ -form w defines a hermitian form h on V : $h(x, y) = w(ix, y) + iw(x, y)$. This is a 1-1 correspondence. The image c_1 is equal to the kernel of the projection $H^2(A; \mathbb{Z}) \hookrightarrow H^2(A; \mathbb{C}) \rightarrow H^{0,2}(A) = H^2(A; \mathcal{O})$, it coincides with integral 2-forms of type $(1, 1)$, which is loosely denoted by $H^{1,1}(A; \mathbb{Z})$. Note that since any constant (p, q) -form on V gives a closed (p, q) -form on A , the Hodge decomposition $H^n(A; \mathbb{C}) \simeq \oplus_{p+q=n} H^{p,q}(A)$ available even without Kähler structure on A . For $w \in H^{1,1}(A; \mathbb{Z})$ we set $\text{Pic}^w(A)$ to be the set of line bundles L over A with $c_1(L) = w$. Clearly $\text{Pic}^w(A)$ is a torsor over $\text{Pic}^0(A)$.

From (4.1.1) it follows that $\text{Pic}^0(A) \simeq H^1(A; \mathcal{O})/H^1(A; \mathbb{Z})$ is itself a complex torus, equal to the quotient of n -dimensional vector space $H^1(A; \mathcal{O})$ by the lattice $H^1(A; \mathbb{Z})$. We have $\dim A = \dim \text{Pic}^0(A)$ and as we will see $\text{Pic}^0(A)$ should be thought as a dual to A . The isomorphism $\bar{V}^\vee/\Gamma^\vee \rightarrow \text{Pic}^0(A)$ can be described explicitly as follows. The natural map

$$H^1(A; \mathbb{R}) = \text{Hom}_{\mathbb{R}}(V, \mathbb{R}) \rightarrow H^1(A; \mathcal{O}) = \text{Hom}_{\mathbb{C}}(\bar{V}, \mathbb{C})$$

sends $\text{Re } \bar{l}$ to $\bar{l}$. The exponential sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{R} \xrightarrow{e^{2\pi i -}} \text{U}(1) \rightarrow 1$ induces a map $H^1(A; \mathbb{R}) \rightarrow H^1(A; \text{U}(1))$ and sends $\text{Re } \bar{l}$ to $e^{2\pi i \text{Re } \bar{l}}$. By the naturality of long exact sequences it follows that the image of $\bar{l} \in H^1(A; \mathcal{O})$ in $\text{Pic}^0(A) = H^1(A; \mathcal{O}^*)$ is holomorphic bundle corresponding to the unique $\text{U}(1)$ -bundle on $A = V/\Gamma$ with monodromy along γ equal to $e^{2\pi i \text{Re } \bar{l}(\gamma)}$.

2. Equivalently any $L \in \text{Pic}(A)$ can be describe by a collection of meromorphic functions $e_\gamma: V \rightarrow \mathbb{C}^\times$ for each $\gamma \in \Gamma$ satisfying the cocycle condition:

$$e_{\gamma_1 + \gamma_2}(x) = e_{\gamma_1}(\gamma_2 \cdot x) e_{\gamma_2}(x),$$

where we use the multiplicative notation for the action of Γ on V . Given the cocycle $e_\gamma(x)$ we define the corresponding line bundle on A as $V \times \mathbb{C}/(x, t) \sim (\gamma \cdot x, e_\gamma(x)t)$. Two line bundles corresponding to cocycles e_γ and e'_γ are isomorphic iff the cocycles differ by a coboundary:

$$e'_\gamma(x) = e_\gamma(x) \cdot \frac{f(\gamma \cdot x)}{f(x)},$$

for some $f \in \mathcal{O}^*(V)$. In other words there is an isomorphism with the group cohomology $\text{Pic}(A) \simeq H^1(\Gamma; \mathcal{O}^*(V))$. Using the exponential short exact sequence again we get an isomorphism of long exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^1(\Gamma; \mathbb{Z}) & \longrightarrow & H^1(\Gamma; \mathcal{O}(V)) & \xrightarrow{e^{2\pi i -}} & H^1(\Gamma; \mathcal{O}^*(V)) & \xrightarrow{c_1} & H^2(\Gamma; \mathbb{Z}) \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & H^1(A; \mathbb{Z}) & \longrightarrow & H^1(A; \mathcal{O}) & \xrightarrow{e^{2\pi i -}} & H^1(A; \mathcal{O}^*) & \xrightarrow{c_1} & H^2(A; \mathbb{Z}) \end{array}$$

3. We are going to describe explicit cocycles which parametrize $\text{Pic}^w(A)$ for the given $w \in H^{1,1}(A; \mathbb{Z})$. Assume α is a map $\Gamma \rightarrow \text{U}(1)$ is such that

$$\frac{\alpha(\gamma_1 + \gamma_2)}{\alpha(\gamma_1)\alpha(\gamma_2)} = e^{i\pi w(\gamma_1, \gamma_2)}.$$

The right hand side takes value ± 1 and the set of such α is a torsor over all homomorphisms $\text{Hom}_{\mathbb{Z}}(\Gamma, \text{U}(1))$. We will show that the pairs (α, w) parametrize $\text{Pic}^w(A)$.

Let $h(x, y) = w(ix, y) + iw(x, y)$ be the hermitian form corresponding to w . Set

$$e_\gamma(x) = \alpha(\gamma) e^{\frac{\pi}{2} h(\gamma, \gamma) + \pi h(x, \gamma)}. \quad (4.1.2)$$

Proposition 4.1.3. *The collection $e_\gamma(x)$ is a cocycle. The line bundle $L_{(\alpha, w)}$ corresponding to the cocycle satisfies $c_1(L) = w$.*

Proof. DIY. □

Note that the set $\{(\alpha, w)\}$ becomes a group if we set $(\alpha, w) \cdot (\alpha', w') = (\alpha\alpha', w + w')$.

Theorem 4.1.4 (Appel-Humbert). *The map $(\alpha, w) \rightsquigarrow L_{(\alpha, w)} \in \text{Pic}^w(A)$ induces an isomorphism $\{(\alpha, w)\} \rightarrow \text{Pic}(A)$.*

In particular, for $w = 0$ we obtain that there is an isomorphism

$$\text{Hom}(\Gamma, \text{U}(1)) \rightarrow \text{Pic}^0(A).$$

In other words, the construction which sends $U(1)$ -flat bundle over A to the corresponding holomorphic line bundle is an isomorphism onto $\text{Pic}^0(A)$.

Exercise Use $\partial\bar{\partial}$ -lemma and show directly that any $L \in \text{Pic}(X)$ such that $c_1(L) \in H^2(X; \mathbb{Z})$ is torsion, where X is a Kähler manifold, admits a unique hermitian metric such that the corresponding Chern connection is flat. It is called hermitian Yang-Mills metric.

This is equivalent to Narasimhan-Seshadri theorem in rank one case. Even in this case it is only one-half of the story: Simpson's correspondence says that there is a bijection between $\text{Hom}(\Gamma, \mathbb{C}^\times)$ and rank one Higgs fields on A .

Exercise Assume X is a Kähler manifold. Show that $H^1(X; \mathbb{C}^\times)$ is isomorphic to the group of flat line bundles. Deduce from the previous exercise that $H^1(X; \mathcal{O}^\times)/H^1(X; \mathbb{C}^\times)$ is isomorphic to the torsion-free group $H^{1,1}(X; \mathbb{Z}) = H^2(X; \mathbb{Z}) \cap H^{1,1}(X)$.

4.2 Correspondences and Poincaré line bundle

Assume $A = V/\Gamma$ and $A' = V'/\Gamma'$ are complex tori.

Definition 4.2.1. A correspondence $P \in \text{Cor}(A, A')$ is a line bundle over $A \times A'$ provided with trivializations over $A \times e'$ and $e \times A'$.

Correspondences form a group under tensor product of line bundles.

Definition 4.2.2. A linear map $f: V \times V' \rightarrow \mathbb{R}$ is *hermitian* if $f(ix, iy) = f(x, y)$.

1. We have the following simple description of correspondences.

Proposition 4.2.3. *There is a bijection between correspondences $P \in \text{Cor}(A, A')$ and hermitian maps $\Gamma \times \Gamma' \rightarrow \mathbb{Z}$.*

Proof. By Appel-Humbert theorem P is given by a form w on $\Gamma \times \Gamma'$ of type $(1, 1)$ and a map $\alpha: \Gamma \times \Gamma' \rightarrow \text{U}(1)$. Let $f(a, a') = w((a, 0), (0, a'))$, this defines a hermitian map $f: \Gamma \times \Gamma' \rightarrow \mathbb{Z}$.

Since P restricts trivially to $e \times A'$ and $A \times e'$ we obtain that $V \times 0$ and $0 \times V'$ are Lagrangian, hence

$$w((a, a'), (b, b')) = f(a, b') - f(b, a').$$

Similarly $\alpha((\gamma, 0)) = 1 = \alpha((0, \gamma'))$, hence

$$\alpha((\gamma, \gamma')) = \alpha((\gamma, 0) + (0, \gamma')) = e^{i\pi f(\gamma, \gamma')}.$$

Clearly the construction is reversible. □

Remark 4.2.4. More precisely $\text{Cor}(A, A')$ is a discrete groupoid equivalent to the set of hermitian maps as above.

Following the notation of the preceding proof, given such f we describe the associated cocycle explicitly. Immediately we have

$$\alpha((\gamma, \gamma')) = e^{i\pi f(\gamma, \gamma')}. \quad (4.2.5)$$

Note that if h is the hermitian form corresponding to w , then

$$h((a, a'), (b, b')) = h(a, b') + \overline{h(b, a')},$$

where $h(a, b') := h((a, 0), (0, b')) = f(ia, b') + if(a, b')$. Then

$$\frac{\pi}{2}h((\gamma, \gamma'), (\gamma, \gamma')) = \pi \operatorname{Re} h(\gamma, \gamma') = \pi f(i\gamma, \gamma'). \quad (4.2.6)$$

Similarly

$$h((a, a'), (\gamma, \gamma')) = h(a, \gamma') + \overline{h(\gamma, a')} = f(ia, \gamma') + if(a, \gamma') + f(i\gamma, a') - if(\gamma, a'). \quad (4.2.7)$$

2. Recall that $H^1(A; \mathbb{Z}) \simeq \Gamma^\vee$ coincides with the space of real 1-forms on V integral over Γ . On the other hand $H^1(A; \mathcal{O})$ is naturally isomorphic to $\bar{V}^\vee$: the space of $(0, 1)$ -forms on V . The lattice $\Gamma^\vee \subset \bar{V}^\vee$ is given by $(0, 1)$ -forms with real part integral on Γ . We have a natural isomorphism of complex tori

$$\operatorname{Pic}^0(A) \simeq \bar{V}^\vee / \Gamma^\vee.$$

Define a hermitian map $f: V \times \bar{V}^\vee \rightarrow \mathbb{R}$ by setting $f(a, \bar{l}) = -\operatorname{Re} \bar{l}(a)$, where $\bar{l}(a) = \overline{l(a)}$ for any $l \in V^\vee$. By definition

$$\bar{l} \in \Gamma^\vee \iff \operatorname{Re} \bar{l}(a) \in \mathbb{Z},$$

for all $a \in \Gamma$. Clearly $f(ia, i\bar{l}) = -f(ia, \bar{l}) = f(a, \bar{l})$. Hence f is hermitian and integral: $f: \Gamma \times \Gamma^\vee \rightarrow \mathbb{Z}$. By the previous this defines a correspondence

$$\mathcal{P} \in \operatorname{Pic}(A \times \operatorname{Pic}^0(A))$$

called *Poincaré line bundle*.

The bundle $\mathcal{P}$ is universal in the following sense. For each $\bar{l} \in \bar{V}^\vee / \Gamma^\vee$ the restriction $\mathcal{P}|_{A \times \bar{l}} \in \operatorname{Pic}(V/\Gamma)$ is isomorphic to the line bundle $\bar{l} \in \bar{V}^\vee / \Gamma^\vee \simeq \operatorname{Pic}^0(V/\Gamma)$ described above.

To see this one has to recover the cocycle of $\mathcal{P}_{A \times \bar{l}}$ by using the preceding Proposition. Namely, let $\gamma \in \Gamma, \bar{\mu} \in \Gamma^\vee$ and $a \in V, \bar{l} \in \bar{V}^\vee$. Using the above formulas one can check that the cocycle of $\mathcal{P}$ is given by the formula

$$e_{(\gamma, \bar{\mu})}(a, \bar{l}) = \exp i\pi(\mu(\gamma \cdot a) + \bar{l}(\gamma)).$$

The restriction $\mathcal{P}|_{A \times \bar{l}}$ corresponds to the cocycle

$$e'_\gamma(a) = e^{i\pi \bar{l}(\gamma)},$$

obtained by setting $\mu = 0$ in the above formula. It is convenient to change the cocycle e'_γ by the coboundary of the holomorphic function $a \rightarrow e^{i\pi l(a)}$ to get an equivalent cocycle e_γ with values in $\operatorname{U}(1)$:

$$e_\gamma(a) = e^{2\pi i \operatorname{Re} \bar{l}(\gamma)}.$$

This computation shows that $\mathcal{P}_{A \times \bar{l}}$ is $\operatorname{U}(1)$ -bundle on A with monodromy along the loop γ equal to $e_\gamma = e^{2\pi i \operatorname{Re} \bar{l}(\gamma)}$.

Now, if $P \in \text{Cor}(A, A')$ is a correspondence, then it induces a morphism $u: A' \rightarrow \text{Pic}^0(A)$ by mapping $x' \in A'$ to $P_{A \times x'} \in \text{Pic}^0(A)$. As we saw above $\text{Pic}^0(A)$ and Poincaré line bundle $\mathcal{P} \in \text{Cor}(A, \text{Pic}^0(A))$ is a universal pair among such: there is a unique $u: A' \rightarrow \text{Pic}^0(A)$ such that the correspondences $u^*\mathcal{P}$ and P are canonically isomorphic. To prove it note that $u^*\mathcal{P} \otimes P^{-1} \in \text{Cor}(A, A')$ has trivial restrictions to all slices $A \times x'$.

Definition 4.2.8. If A is a complex tori, then a dual complex tori is a pair of a complex torus $A^\vee$ and a correspondence $P \in \text{Cor}(A, A^\vee)$, such that the induced morphism $A^\vee \rightarrow \text{Pic}^0(A)$ is an isomorphism.

We saw that $A^\vee \rightarrow \text{Pic}^0(A)$ is an isomorphism $\iff$ the hermitian map $\Gamma \times \Gamma^\vee \mathbb{Z}$ underlying P is a perfect pairing. Hence the notion of duality is symmetric: the swapping coordinates allows to write $P \in \text{Cor}(A^\vee, A)$, thus $A \simeq (A^\vee)^\vee$. So we can refer to $\text{Pic}^0(A)$ and the Poincaré line bundle is *the* dual to A . It follows that there is a natural isomorphism $A \simeq \text{Pic}^0(\text{Pic}^0(A))$.

Note also that any morphism $f: A' \rightarrow A^\vee$ induces the dual $f^\vee: A \rightarrow A'^\vee$, such that $(f^\vee)^\vee = f$. The maps $A'^\vee \times A' \xleftarrow{(f^\vee, \text{id})} A \times A' \xrightarrow{(\text{id}, f)} A \times A^\vee$ are consistent in the sense that $(\text{id}, f)^*\mathcal{P}$ and $(f^\vee, \text{id})^*\mathcal{P}'$ are naturally isomorphic.

4.3 Some algebraic-geometry

Recall that abelian varieties are sufficiently rigid objects and any morphism preserving the neutral element is a homomorphism. More precisely we have the following theorem.

Theorem 4.3.1. *Any morphism of $A \rightarrow B$ of abelian varieties considered as varieties is the composition of a shift in B and a homomorphism.*

Proposition 4.3.2. *Assume the morphism $p: E \rightarrow S$ is smooth projective and $L \in \text{Pic}(E)$ is such that $L|_{p^{-1}(s)}$ is trivial for any $s \in S$. Then the natural morphism $p^*p_*L \simeq L$ is an isomorphism. In other words L is a pullback of the line bundle $p_*L \in \text{Pic}(S)$.*

Theorem 4.3.3 (Square theorem).

Theorem 4.3.4 (Cube theorem). *Assume $X \times Y \times Z$ is a product of connected projective varieties with point $x_0 \times y_0 \times z_0$ and $L \in \text{Pic}(X \times Y \times Z)$ is such that the restrictions $L|_{X \times Y \times z_0}$, $L|_{X \times y_0 \times Z}$ and $L|_{x_0 \times Y \times Z}$ are trivial. Then L is trivial.*

Remark 4.3.5. In a certain sense the theorem says that Pic is a polynomial functor of degree ≤ 2 .

As a corollary assume A is an abelian variety and $L \in \text{Pic}(A)$. Let $M \in \text{Pic}(A \times A \times A)$ be defined on the fibers as follows

$$M_{x,y,z} = L_{x+y+z} \otimes L_{x+y}^{-1} \otimes L_{y+z}^{-1} \otimes L_{x+z}^{-1} \otimes L_x \otimes L_y \otimes L_z.$$

Clearly $M_{e,*,*}, M_{*,e,*}, M_{*,*,e}$ are trivial bundles, hence M is trivial.

We have an immediate corollary.

Corollary 4.3.6. *Assume $f, g, h: M \rightarrow A$ are morphisms from a variety M to the abelian variety A . A map $\text{Pic}(A) \rightarrow \text{Pic}(M)$ given by*

$$(f^* + g^* + h^*) - (f + g)^* - (g + h)^* - (f + h)^* + f^* + g^* + h^*$$

is zero.

Let $[n]: A \rightarrow A$ denotes the multiplication by $n \in \mathbb{Z}$. If $n \neq 0$, then $[n]$ is an isogeny of degree $|n|^{2 \dim A}$: $(\mathbb{Z}/n)^{2 \dim A} \simeq \ker[n] \subset A$.

Corollary 4.3.7. *For any $L \in \text{Pic}^w(A)$ we have*

$$[n]^* L \simeq L^{\otimes \binom{n+1}{2}} \otimes [-1]^* L^{\otimes \binom{n-1}{2}} \in \text{Pic}^{n^2 \cdot w}(A),$$

moreover if $L \in \text{Pic}^0(A)$, then

$$[n]^* L \simeq L^{\otimes n}.$$

Proof. 1. For the first assertion use the previous corollary to $f = [n+1], g = [1], h = [-1]$ and solve the recurrent relation.

2. Note that if $f: A \rightarrow A'$ is morphism of complex tori and $f^*: \text{Pic}^0(A') \rightarrow \text{Pic}^0(A)$ is the induced map, then its differential at identity $df^*: H^1(A'; \mathcal{O}) \rightarrow H^1(A; \mathcal{O})$ is equal to the induced map of cohomology. It follows that the map $[n]: A \rightarrow A$ induces the differential $d[n]^*$ equal to the multiplication by n . It remains to note that a map of complex tori is determined by its differential at identity, hence $[n]^*: \text{Pic}^0(A) \rightarrow \text{Pic}^0(A)$ is the multiplication by n . $\square$

Let $m: A \times A \rightarrow A$ denotes the group operation. Denote by $m_x: A \rightarrow A$ the shift by $x \in A$, so $m_x \circ m_y = m_{x+y}$. Let $x \in A$ and $L \in \text{Pic}(A)$, define

$$\Phi(x, L) = m_x^*(L) \otimes L^{-1}.$$

Note that $\Phi(x, -): \text{Pic}(A) \rightarrow \text{Pic}(A)$ preserves the trivial line bundle, hence it is a homomorphism. The same holds for $\Phi(-, L): A \rightarrow \text{Pic}(A)$. This implies an isomorphism

$$m_{x+y}^* L \simeq m_x^* L \otimes m_y^* L \otimes L^{-1},$$

which follows also from the cube theorem, because it says that for given $x, y \in A$ the bundle

$$L_{x+y+z} \otimes L_{y+z}^{-1} \otimes L_{x+z}^{-1} \otimes L_z$$

in $z \in A$ is trivial. It follows that $\Phi(x, \Phi(y, L))$ is always trivial:

$$\Phi(x, \Phi(y, L)) = \Phi(x, m_y^* L \otimes L^{-1}) = \Phi(x, m_y^* L) \otimes \Phi(x, L)^{-1} = m_{x+y}^* L \otimes m_y^* L^{-1} \otimes m_x^* L^{-1} \otimes L.$$

Let $A = V/\Gamma$.

Proposition 4.3.8. *Assume $L \in \text{Pic}^w(A)$ for some $w \in H^{1,1}(A; \mathbb{Z})$. The image of*

$$\Phi(-, L): A \rightarrow \text{Pic}(A)$$

is a closed connected subgroup in $\text{Pic}^0(A)$. The differential

$$d\Phi(-, L): T_e A \rightarrow T_e \text{Pic}^0(A),$$

after the identification with a map $V \rightarrow \overline{V}^\vee$, is equal to w . In particular, the form w is non-degenerate if and only if $\Phi(-, L)$ is an isogeny onto $\text{Pic}^0(A)$, which has degree $|\det w|$.

Proof. The first assertion is clear. Let $A = V/\Gamma$. Assume $L \in \text{Pic}^w(A)$ is given by the cocycle $e_\gamma(x)$ of the form (4.1.2) for $\gamma \in \Gamma$, $x \in V$ and integral form $w: \Gamma \times \Gamma \rightarrow \mathbb{Z}$ corresponding to a hermitian form h . Then

$$\frac{e_\gamma(x+a)}{e_\gamma(x)} = e^{\pi h(a, \gamma)}$$

is a cocycle representing $\Phi(a, L) \in H^1(\Gamma; \mathcal{O}^*(V)) \simeq \text{Pic}(A)$. Multiplying by the coboundary $\gamma \rightsquigarrow e^{l(x+\gamma)-l(x)} = e^{l(\gamma)}$ of a holomorphic function, where $l(x) = -\pi h(a, x)$, $x \in V$, we obtain an equivalent cocycle

$$e^{\pi h(a, \gamma) - \pi \overline{h(a, \gamma)}} = e^{i\pi \text{Im } h(a, \gamma)} = e^{2\pi i w(a, \gamma)}.$$

Thus the line bundle $\Phi(a, L)$ is given by the cocycle $\gamma \rightsquigarrow e^{2\pi i w(a, \gamma)}$ with constant values in $\text{U}(1)$. By Appel-Humbert theorem this cocycle represents a trivial class iff $a \in \ker w$. In particular $\ker w \subset V$ is the tangent space to the subgroup $\ker \Phi(-, L)$. $\square$

4.4 Ample line bundles, positivity and polarizations

1. Recall that for a line bundle L over a complex manifold X one attempts to define a natural map $X \rightarrow \mathbb{P}(H^0(X; L)^\vee)$ which sends $x \in X$ to the hyperplane of sections vanishing at x . Explicitly, if $e_0, \dots, e_n$ is a basis of $H^0(X; L)$, then the corresponding map can be described by the formula $x \rightsquigarrow [e_0(x) : \dots : e_n(x)] \in \mathbb{P}^n$. The map $X \rightarrow \mathbb{P}(H^0(X; L)^\vee)$ is not well defined at the *base points* $x \in X$, where all sections vanish.

Definition 4.4.1. The line bundle $L \in \text{Pic}(X)$ is *ample* if the canonical map

$$X \rightarrow \mathbb{P}(H^0(X; L^{\otimes k})^\vee)$$

is an embedding for some $k > 0$.

In short L is ample when its power $L^{\otimes k}$ for sufficiently large k admits enough sections to separate all points in X .

Exercise Check that if the map $f: X \rightarrow \mathbb{P}^n = \mathbb{P}(H^0(X; L)^\vee)$ is well-defined, then one has a natural isomorphism $f^*(\mathcal{O}_{\mathbb{P}^n}(1)) \xrightarrow{\sim} L$. Conversely, if $i: X \subset \mathbb{P}^n$, then the canonical embedding corresponding to the bundle $L = i^*\mathcal{O}_{\mathbb{P}^n}(1)$ is naturally isomorphic to $i \iff$ the restriction map $H^0(\mathbb{P}^n; \mathcal{O}(1)) \rightarrow H^0(X; L)$ is an isomorphism.

We call a $(1, 1)$ -form w positive if $w(ix, x) > 0$.

Theorem 4.4.2 (Kodaira). *A line bundle L over compact complex manifold X is ample $\iff$ its Chern class $c_1(L)$ admits a positive representative: there is 2-form w of type $(1, 1)$ such that $[w] = c_1(L)$ and w is pointwise positive.*

In particular the ampleness of L is the property of $c_1(L)$ and hence $L \otimes M$ is ample for all $M \in \text{Pic}^0(X)$.

2. Usually an abelian variety A is defined as a smooth projective algebraic group, but without specifying the projective embedding. It is well-known that A is always commutative, so it is topologically a torus, hence it is a quotient of $H_1(A; \mathbb{R})/H_1(A; \mathbb{Z})$. To see the complex structure one can consider the isomorphism $A \xrightarrow{\sim} H^0(A; \Omega^1)^\vee / H_1(A; \mathbb{Z})$ given by integration on holomorphic differentials.

Assume $A = V/\Gamma$ is a complex torus. By Appel-Humbert theorem applied to $L \in \text{Pic}^w(A)$, $c_1(L)$ is represented by a $(1, 1)$ -form w on Γ . Hence Kodaira theorem implies that L is ample $\iff w \in H^{1,1}(A; \mathbb{Z})$ is positive.

Definition 4.4.3. A polarized abelian variety (A, w) is a complex torus $A = V/\Gamma$ provided with a positive integral form w on Γ .

Remark 4.4.4. In dimension $n \geq 2$ not every complex torus admit a non trivial line bundle L with $c_1(L) \neq 0$, let alone a polarization. General lattice $\Gamma \subset \mathbb{C}^n$ doesn't admit any $(1,1)$ -form w , so $H^{1,1}(V/\Gamma; \mathbb{Z}) \simeq \pi_0(\text{Pic}(V/\Gamma))$ can be trivial. In general the *Picard number* $r = \text{rk } \pi_0(\text{Pic}(V/\Gamma))$ can take any value in $0 \leq r \leq \binom{n}{2}$.

We call a correspondence $P \in \text{Cor}(A, A)$ *symmetric* if it is unchanged by swapping two factors. Recall that $m: A \times A \rightarrow A$ is the multiplication, $p_{1,2}: A \times A \rightarrow A$ are projections and $\Delta: A \rightarrow A \times A$ is the diagonal map.

Proposition 4.4.5. *Any symmetric correspondence $P \in \text{Cor}(A, A)$ has the form*

$$P = m^*L \otimes p_1^*L^{-1} \otimes p_2^*L^{-1},$$

where $L = \Delta^*P$. If $L \in \text{Pic}^w(A)$, then P corresponds to the hermitian map $w: \Gamma \times \Gamma \rightarrow \mathbb{Z}$.

Proof. By 4.2.3, a correspondence $P \in \text{Cor}(A, A)$ is a hermitian map

$$w: \Gamma \times \Gamma \rightarrow \mathbb{Z}.$$

Recall that w defines an integral $(1,1)$ -form on $V \times V$ by

$$v((a, a'), (b, b')) = w(a, b') - w(b, a').$$

The correspondence P is symmetric $\iff v((a, a'), (b, b')) = v((b, b'), (a, a'))$. In other words a symmetric correspondence is equivalent to a skew-symmetric hermitian map $w: \Gamma \times \Gamma \rightarrow \mathbb{Z}$.

Given $L \in \text{Pic}^w(A)$, the correspondence $P' = m^*L \otimes p_1^*L^{-1} \otimes p_2^*L^{-1} \in \text{Pic}(A \times A)$ has the underlying $(1,1)$ -form equal to $v = m^*w - p_1^*w - p_2^*w$, where $m: V \times V \rightarrow V$ is the sum and p_i are the corresponding projections. Hence

$$v((a, a'), (b, b')) = w(a + a', b + b') - w(a, b) - w(a', b) = w(a, b') - w(b, a')$$

and thus the hermitian map of P' is equal to $w: \Gamma \times \Gamma \rightarrow \mathbb{Z}$.

On the other hand $\Delta^*P' = [2]^*L \otimes L^{-2} = L$. It follows that P and P' are equal. $\square$

Corollary 4.4.6. *A polarization (A, w) is a symmetric correspondence $P \in \text{Cor}(A, A)$ such that $\Delta^*P \in \text{Pic}(A)$ is ample.*

By Proposition 4.3.8, the polarization can be thought as a pair (A, L) , where L is an ample line bundle defined up to multiplication by $\text{Pic}^0(A)$. Then it provides an isogeny

$$\Phi(-, L): A \rightarrow A^\vee \simeq \text{Pic}^0(A).$$

We call the isogeny *symmetric* if $\Phi(-, L)^\vee: A \simeq (A^\vee)^\vee \rightarrow A^\vee$ is equal to $\Phi(-, L)$.

Proposition 4.4.7. *A non degenerate form $w \in H^{1,1}(A)$ is equivalent to a symmetric isogeny $A \rightarrow A^\vee$ of degree $|\det w|$.*

Proof. 1. Consider a map $m_x: A \rightarrow A$ given by multiplication by $x \in A$. If A is polarized by an ample $L \in \text{Pic}^w(A)$, then $L \rightsquigarrow m_x^*L \otimes L^{-1}$ gives a map $\Phi(-, L): A \rightarrow \text{Pic}^0(A)$. By Proposition 4.3.8, $\Phi(-, L)$ is an isogeny which depends only on w .

2. Given an isogeny $f: A \rightarrow A^\vee$, the pullback of Poincaré line bundle $\mathcal{P} \in \text{Cor}(A^\vee, A)$ provides a correspondence $f^*\mathcal{P} \in \text{Cor}(A, A)$. If f was symmetric, then so is $f^*\mathcal{P}$. $\square$

Hence a polarization is a symmetric isogeny $A \rightarrow A^\vee \simeq \text{Pic}^0(A)$ of the form $x \rightsquigarrow m_x^*L \otimes L^{-1}$ for an ample $L \in \text{Pic}^w(A)$.

4.5 Principal polarizations and Siegel's upper half-space

Definition 4.5.1. A *principal polarization* is a polarization (A, w) , where $|\det w| = 1$.

In other words the induced isogeny $A \rightarrow A^\vee$ is an isomorphism equal to $\Phi(-, L)$ for an ample line bundle $L \in \text{Pic}^w(A)$, where $w: \Gamma \times \Gamma \rightarrow \mathbb{Z}$ is a perfect pairing.

Our main example of principally polarized abelian varieties is given by the Jacobian of a curve. Namely, let Σ be a complex curve of genus g . The lattice $H^1(C; \mathbb{Z}) \subset H^1(\Sigma; \mathcal{O})$ admits a natural $(1, 1)$ -form given by the cup product followed by the integration over $[\Sigma]$. The cup pairing is perfect. It follows that the complex torus $A = \text{Pic}^0(\Sigma) \simeq H^1(\Sigma; \mathcal{O})/H^1(C; \mathbb{Z})$ is a principally polarized abelian variety.

Consider $\mathbb{R}^{2n}$ with the standard symplectic form w and $\mathbb{C}^n$ with the standard complex structure. Start with the following observation. Denote by (V, Γ, ω) a complex vector space V of dimension n with an embedding of lattice $\mathbb{Z}^{2n} \hookrightarrow V$ and $(1, 1)$ -form ω such that $\omega|_{\mathbb{Z}^{2n}}$ is the standard integral symplectic form. Then the groupoid of such objects $\{(V, \mathbb{Z}^{2n}, \omega)\}$ is discrete, i.e. the automorphism group of $(V, \mathbb{Z}^{2n}, \omega)$ is trivial and one wants to understand the set $\{V, \mathbb{Z}^{2n}, \omega\}/\text{iso}$. It enjoys a natural action $\text{Sp}(2n, \mathbb{R})$ such that the subgroup $\text{Sp}(2n, \mathbb{Z})$ acts via the precomposition of the lattice embedding.

The set of isomorphism classes $\{(V, \mathbb{Z}^{2n}, \omega)\}/\text{iso}$, admit two equivalent descriptions.

1. The first is given by the set of complex structures J on $\mathbb{R}^{2n}$ such that the standard form w has type $(1, 1)$. The action of $\text{Sp}(2n, \mathbb{R})$ is given by $J \rightsquigarrow aJa^{-1}$ for $a \in \text{Sp}(2n, \mathbb{R})$. Recall the set of all complex structures is identified with $\text{GL}(2n, \mathbb{R})/\text{GL}(n, \mathbb{C})$ via the map which sends $g \in \text{GL}(2n, \mathbb{R})$ to gIg^{-1} , where I is the standard complex structure on $\mathbb{R}^{2n}$. In particular, the stabilizer of I is equal to

$$\{a \in \text{Sp}(2n, \mathbb{R}) \mid aJa^{-1} = J \in \text{GL}(2n, \mathbb{R})/\text{GL}(n, \mathbb{C})\} = \text{U}(n).$$

Note that the pair (w, J) such that J is a complex structure and w is positive of type $(1, 1)$ with respect to J is equivalent to a positive hermitian structure on $\mathbb{R}^{2n}$. The existence of orthogonal bases implies that any two hermitian vector spaces of the same dimension are unitary isomorphic. An isomorphism is unitary if it intertwines the complex and symplectic structures. In particular:

$$\text{GL}(\mathbb{R}^{2n})/\text{GL}(\mathbb{R}^{2n}, I) \simeq \{\text{complex structure } J \text{ on } \mathbb{R}^{2n}\},$$

$$\text{Sp}(\mathbb{R}^{2n})/\text{Sp}(\mathbb{R}^{2n}) \cap \text{GL}(\mathbb{R}^{2n}, I) \simeq \{\text{complex structure } J \text{ on } \mathbb{R}^{2n} \text{ s.t. } w(J-, J-) > 0\}. \quad (4.5.2)$$

2. The second description is given by the set of lattice embeddings $i: \mathbb{Z}^{2n} \hookrightarrow \mathbb{C}^n$ such that (1) the last n basis elements map to the basis of $\mathbb{C}^n$ and (2) the induced 2-form $i_*(w)$ on $\mathbb{C}^n$ has type $(1, 1)$. The first condition reflects the fact that we are interested in lattices up to the left action by $\text{GL}(n, \mathbb{C})$. The lattice embedding can be written as $(Z|\text{id}_n)$, where $Z \in \text{Mat}_{n \times n}(\mathbb{C})$ equal to $X + iY$ for some real matrices X, Y with invertible Y . Thus the right action by $\text{Sp}(2n, \mathbb{R})$ is given by the formula

$$(Z|\text{id}) \cdot \begin{pmatrix} A & B \\ C & D \end{pmatrix} = (ZA + C|ZB + D) \sim ((ZB + D)^{-1}(ZA + C)|\text{id}),$$

where we used $\text{GL}(n, \mathbb{C})$ action to identify the second component with id . One can use transposition in order to obtain more recognizable left action:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \cdot \begin{pmatrix} Z^* \\ 1 \end{pmatrix} = (AZ^* + B)(CZ^* + D)^{-1}.$$

Here we use the agreement that $\begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathrm{Sp}(2n, \mathbb{R})$, if and only if

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} 0 & -\mathrm{id} \\ \mathrm{id} & 0 \end{pmatrix} \begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix} = \begin{pmatrix} 0 & -\mathrm{id} \\ \mathrm{id} & 0 \end{pmatrix},$$

or

$$BA^T = AB^T, \quad DC^T = CD^D, \quad AD^T - BC^T = \mathrm{id}. \quad (4.5.3)$$

In order to agree with the literature it might be better to compare the results in terms of Z^* . Our goal is to express the conditions on the induced form on $\mathbb{C}^n$ in terms of $Z \in \mathrm{Mat}_{n \times n}(\mathbb{C})$.

We can write

$$(Z|\mathrm{id}) = \begin{pmatrix} X & \mathrm{id} \\ Y & 0 \end{pmatrix}$$

and its inverse

$$\begin{pmatrix} X & \mathrm{id} \\ Y & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & Y^{-1} \\ \mathrm{id} & -XY^{-1} \end{pmatrix}.$$

Let $X' = Y^{-1}$ and $Y' = -XY^{-1}$. The matrix of the standard 2-form on $\mathbb{R}^{2n}$ is $W = \begin{pmatrix} 0 & -\mathrm{id} \\ \mathrm{id} & 0 \end{pmatrix}$

and that of the complex structure $I = \begin{pmatrix} 0 & -\mathrm{id} \\ \mathrm{id} & 0 \end{pmatrix}$. The induced form $i_*(w)$ in the basis given by standard identification $\mathbb{C}^n \simeq \mathbb{R}^n \oplus I\mathbb{R}^n$, has the following matrix

$$\begin{pmatrix} 0 & \mathrm{id} \\ X'^T & Y'^T \end{pmatrix} \begin{pmatrix} 0 & -\mathrm{id} \\ \mathrm{id} & 0 \end{pmatrix} \begin{pmatrix} 0 & X' \\ \mathrm{id} & Y' \end{pmatrix},$$

which is equal to

$$W' := \begin{pmatrix} 0 & X' \\ -X'^T & Y'^T X' - X'^T Y' \end{pmatrix}.$$

The condition that W' has type $(1, 1)$ is $-IW'I = W'$. The positivity of W' is equivalent to $-IW' > 0$ (positive definite). One checks that the first condition is equivalent to X', Y' and hence X, Y to be symmetric matrices. The second condition is equivalent to $-Y > 0$. The convex complex manifold of all symmetric complex matrices $Z^* \in \mathrm{Mat}_{n \times n}(\mathbb{C})$ such that the imaginary part of Z^* is positive definite, is called *Siegel's upper halfspace* $\mathcal{H}_n$. We have $\dim \mathcal{H}_n = n(n+1)/2$. By the above we obtain another description of (4.5.2)

$$\mathcal{H}_n = \{X + iY \mid X^T = X, Y^T = Y, Y > 0\} = \{\text{complex structure } J \text{ on } \mathbb{R}^{2n} \text{ s.t. } w(J-, -) > 0\}, \quad (4.5.4)$$

where the condition $w(J-, -) > 0$ means that w is positive $(1, 1)$ -form with respect to J . Given $Z^* = X - iY \in \mathcal{H}_n$ the form $i_*(W)$ depends only on the imaginary part:

$$W' = \begin{pmatrix} 0 & Y^{-1} \\ -Y^{-1} & 0 \end{pmatrix}.$$

Proposition 4.5.5. *The group $\mathrm{Sp}(2n, \mathbb{R})$ acts on $\mathcal{H}_n$ transitively, one has an isomorphism of $\mathrm{Sp}(2n, \mathbb{R})$ -spaces*

$$\mathrm{Sp}(2n, \mathbb{R})/\mathrm{U}(n) \simeq \mathcal{H}_n.$$

Proof. We already saw that the stabilizer of $i \in \mathcal{H}_n$ with respect to the action of $\mathrm{Sp}(2n, \mathbb{R})$ is equal to $\mathrm{U}(n) = \mathrm{GL}(2n, \mathbb{C}) \cap \mathrm{Sp}(2n, \mathbb{R})$. Moreover $\dim_{\mathbb{R}} \mathrm{Sp}(2n, \mathbb{R}) = n(2n+1)$ and $\dim_{\mathbb{R}} \mathrm{U}(n) = n^2$, hence $\mathrm{Sp}(2n/\mathbb{R})/\mathrm{U}(n)$ is a submanifold of real dimension $n(n+1)$ in $\mathcal{H}_n$. More directly one

checks that for any $X + iY \in \mathcal{H}_n$ one can set $C = 0, B = XA^{-1}, D = A^{-1}$ for a symmetric A such that $A^2 = Y$. Then the formulas above show that $\begin{pmatrix} A & XA^{-1} \\ 0 & A^{-1} \end{pmatrix} \in \mathrm{Sp}(2n, \mathbb{R})$ and applied to $i \in \mathcal{H}_n$ gives $X + iY$. $\square$

Theorem 4.5.6. *The stack of principally polarized abelian varieties of dimension n is isomorphic to the quotient $\mathrm{Sp}(2n, \mathbb{Z}) \backslash \mathcal{H}_n$.*

Proof. Above we have shown that the moduli of principally polarized abelian varieties A with a chosen standard symplectic basis $\mathbb{Z}^{2n} \simeq H_1(A; \mathbb{Z})$ is isomorphic to the complex manifold $\mathcal{H}_n$. The action by $\mathrm{Sp}(2n, \mathbb{Z})$ simply change the standard symplectic basis and the quotient describes the moduli stack of principally polarized abelian varieties. $\square$

5 Deformations and the geometric quantization

5.1 Deformations of complex structures

A complex manifold M can be described in two ways. The first consists of a pair (M, I) , where $I: TM \rightarrow TM$ is an integrable almost complex structure, such that $I^2 = -1$ and the commutator of vector fields satisfies

$$[T^{1,0}M, T^{1,0}M] \subset T^{1,0}M.$$

The latter is equivalent to the condition that the $(0, 1)$ component of de Rham differential, denoted by d'' , squares to zero: $d''^2: \Lambda^0(M) \rightarrow \Lambda^{0,2}(M)$ vanish. The equation $d''f = 0$ defines the sheaf of holomorphic functions and makes M a smooth manifold with a sheaf of rings over $\mathbb{C}$. The integrability of I is equivalent to the statement that locally M , as a ringed manifold, is isomorphic to a ball in $(\mathbb{C}^n, I)$ equipped with the sheaf of holomorphic functions.

1. Recall that a complex structure on a real vector space (T, I) is equivalent to the decomposition $T \otimes \mathbb{C} = T' \oplus T''$ such that $\overline{T'} = T''$. In our notation $T' = T^{1,0}$ and $T'' = T^{0,1}$. The complex structure I_t for some parameter $t \in B$ such that $I_0 = I$, is a decomposition

$$T \otimes \mathbb{C} = T'_t \oplus T''_t, \quad \overline{T'_t} = T''_t.$$

For small t , T'_t is a graph of a map $\bar{\mu}: T' \rightarrow T''$ and the projection $p'_t: T'_t \rightarrow T'$ is an isomorphism. Thus any vector in T'_t has a form $x' + \bar{\mu}(x')$ for some $x' \in T'$, so we can write $p'^{-1}_t(x') = x' + \bar{\mu}(x')$. Likewise any vector T''_t has a form $x'' + \mu(x'')$ for some $x'' \in T''$, where $\mu: T'' \rightarrow T'$ is the conjugation of $\bar{\mu}$.

The columns of the matrix $\begin{pmatrix} 1 & \mu \\ \bar{\mu} & 1 \end{pmatrix}$, viewed as a map $T' \oplus T'' \rightarrow T' \oplus T''$, span T'_t and T''_t , respectively. The complex structure I_t has a matrix expression

$$I_t = \begin{pmatrix} 1 & \mu \\ \bar{\mu} & 1 \end{pmatrix} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \begin{pmatrix} 1 & \mu \\ \bar{\mu} & 1 \end{pmatrix}^{-1}.$$

We have

$$\begin{pmatrix} 1 & \mu \\ \bar{\mu} & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -\mu \\ -\bar{\mu} & 1 \end{pmatrix} \begin{pmatrix} (1 - \mu\bar{\mu})^{-1} & 0 \\ 0 & (1 - \bar{\mu}\mu)^{-1} \end{pmatrix},$$

and

$$I_t = i \begin{pmatrix} 1 + \mu\bar{\mu} & -2\mu \\ 2\bar{\mu} & -(1 + \bar{\mu}\mu) \end{pmatrix} \begin{pmatrix} (1 - \mu\bar{\mu})^{-1} & 0 \\ 0 & (1 - \bar{\mu}\mu)^{-1} \end{pmatrix}.$$

Let us note the quadratic expansion:

$$I_t = I + 2i \begin{pmatrix} \mu\bar{\mu} & -\mu \\ \bar{\mu} & -\bar{\mu}\mu \end{pmatrix} + o(t^2). \quad (5.1.1)$$

Dually, the decomposition $\Lambda \otimes \mathbb{C} = \Lambda'_t \oplus \Lambda''_t$ is given by the inverse of

$$\begin{pmatrix} 1 & \mu \\ \bar{\mu} & 1 \end{pmatrix}^T = \begin{pmatrix} 1 & \bar{\mu}^T \\ \mu^T & 1 \end{pmatrix},$$

and is equal to

$$\begin{pmatrix} (1 - \bar{\mu}^T \mu^T)^{-1} & 0 \\ 0 & (1 - \mu^T \bar{\mu}^T)^{-1} \end{pmatrix} \begin{pmatrix} 1 & -\bar{\mu}^T \\ -\mu^T & 1 \end{pmatrix},$$

where $\mu^T : \Lambda' \rightarrow \Lambda''$ is the dual to $\mu : T'' \rightarrow T'$. Note that μ and μ^T represent the same element in $\Lambda^{0,1} \otimes T'$, we distinct them only for convenience.

2. Assume (M, I) is a complex manifold. Denote by T and its dual Λ the tangent and cotangent bundle respectively. The almost complex deformation I_t of I is given by $\mu : T'' \rightarrow T'$. We will identify μ and μ^T with an element $\tau \in \Lambda^{0,1} \otimes T'$.

The structure I_t is integrable $\iff [T''_t, T''_t] \subset T''_t$. Locally $\tau = d\bar{z}^i \otimes \tau_i$ for some $\tau_i \in T'$ and the integrability is equivalent to the vanishing of

$$[\bar{\partial}_i + \tau_i, \bar{\partial}_j + \tau_j] = \bar{\partial}_i(\tau_j) - \bar{\partial}_j(\tau_i) + [\tau_i, \tau_j].$$

Invariantly the latter is equivalent to vanishing of Frölicher–Nijenhuis tensor

$$d''\tau + \frac{1}{2}[\tau, \tau] = 0 \in \Lambda^{0,2} \otimes T'. \quad (5.1.2)$$

Here the bracket is defined in the usual way: $[\tau, -]$ is the Lie derivative of degree 1. Namely,

$$[d\bar{z}^i \tau_i, -] = d\bar{z}^i [\tau_i, -] = d\bar{z}^i \mathcal{L}_{\tau_i}(-).$$

For instance

$$[\tau, \tau] = [d\bar{z}^i \tau_i, d\bar{z}^j \tau_j] = [\tau_i, \tau_j] d\bar{z}^i \wedge d\bar{z}^j,$$

since $[\tau_i, d\bar{z}^j] = 0$. Hence I_t is integrable $\iff \tau$ satisfies the Maurer-Cartan equation (5.1.2).

Given a function f , let

$$u' = p'_t(d'_t f), \quad v'' = p''_t(d''_t f),$$

or

$$d'_t f = u' - \tau(u'), \quad d''_t f = v'' - \tau(v'').$$

Since $df = d'f + d''f = d'_t f + d''_t f$, we have

$$\begin{pmatrix} 1 & -\bar{\tau} \\ -\tau & 1 \end{pmatrix} \begin{pmatrix} u' \\ v'' \end{pmatrix} = \begin{pmatrix} d'f \\ d''f \end{pmatrix},$$

or

$$\begin{pmatrix} u' \\ v'' \end{pmatrix} = \begin{pmatrix} (1 - \bar{\tau}\tau)^{-1} & 0 \\ 0 & (1 - \tau\bar{\tau})^{-1} \end{pmatrix} \begin{pmatrix} 1 & \bar{\tau} \\ \tau & 1 \end{pmatrix} \begin{pmatrix} d'f \\ d''f \end{pmatrix}.$$

Thus $v'' = (1 - \tau\bar{\tau})^{-1}(d''f + [\tau, f])$. The function f is holomorphic with respect to I_t , if and only if, $d''f + [\tau, f] = 0$. We obtain a commutative diagram

$$\begin{array}{ccccc} \Lambda^0 & \xrightarrow{\text{id}} & \Lambda^0 & & \\ d'' + [\tau, -] \downarrow & & \downarrow d_t'' & & \\ \Lambda^{0,1} & \xrightarrow{(1-\tau\bar{\tau})^{-1}} & \Lambda^{0,1} & \xrightarrow{p_t''^{-1}} & \Lambda_t^{0,1} \end{array} \quad (5.1.3)$$

The bottom composition naturally extends to an isomorphism of graded algebras over $\Lambda^0(M)$:

$$\Lambda^{0,*}(M) \rightarrow \Lambda_t^{0,*}(M).$$

We already saw that τ acts on $\Lambda^{0,*} \otimes T'$ by a Lie derivation $[\tau, -]$ of degree 1. Locally $\tau = d\bar{z}^i \otimes \tau_i$. In particular $[\tau, f] = d\bar{z}^i[\tau_i, f]$ and $[\tau, d\bar{z}^k] = 0$, the Leibniz rule completely determines $\mathcal{L}_\tau$.

Thus we obtain a graded commutative algebra with a derivation of degree 1 over $\Lambda^0(M)$:

$$(\Lambda^{0,*}(M), d'' + [\tau, -]).$$

We can reinterpret the integrability of I_t as a deformation of Dolbeault complex cdga.

Proposition 5.1.4. *The deformation I_t given by $\tau \in \Lambda^{0,1} \otimes T'$ is integrable, if and only if, $d''\tau + \frac{1}{2}[\tau, \tau] = 0$, i.e. $(\Lambda^{0,*}(M), d'' + [\tau, -])$ is a cdga.*

5.2 Deformation of Kähler polarization after Hitchin

See [ADW91] and [Hit90].

1. Assume (M, I, w) is a Kähler manifold with holomorphic $U(1)$ -bundle L such that the curvature of the Chern connection ∇ on L is equal to iw . We are interested in deformations I_t of I with respect to w , i.e. w is of type $(1, 1)$ with respect to I_t . We think of the deformation I_t in terms of a trivial family of C^∞ -manifolds $M \times B$ over some parameter space B , so $t \in B$ is a point. It follows that such I_t naturally induces a holomorphic structure on L : by the definition $\nabla'' = \frac{1}{2}(1 + iI_t)\nabla$ and we have $\nabla''^2 = 0$ because $[\nabla, \nabla] = iw$ is of type $(1, 1)$ with respect to I_t .

We will focus on first order deformation of I , i.e. $I_t = I + \delta I + o(t)$. According to (5.1.1) we can write $\delta I = 2i(\bar{\tau} - \tau)$ for some $\tau \in \Lambda'' \otimes T'$ such that $d''\tau = 0$. Then

$$\delta(\bar{\partial}) = \frac{1}{2}i \cdot \delta I \circ d = \tau - \bar{\tau},$$

is a derivative of degree 1. Similarly we have $\delta(\nabla'') = \nabla_\tau - \nabla_{\bar{\tau}}$.

Suppose $s \in L$ is a local holomorphic section, i.e. $\nabla''(s) = 0$. Consider the problem of extension of s to a holomorphic section $s + \delta s$. In other words we are interested in solutions of the equation

$$\delta(\nabla''s) = \delta(\nabla'')s + \nabla''\delta s = \nabla'_\tau s + \nabla''\delta s = 0 \in \Lambda^{0,1} \otimes L. \quad (5.2.1)$$

The consistency of the equation, i.e. the existence of local solutions, is equivalent to

$$0 = \nabla''(\nabla'_\tau s) = [\nabla'', \nabla'_\tau](s) = -\iota_\tau[\nabla'', \nabla'](s) = i/2 \cdot \iota_\tau w \cdot s \in \Lambda^{0,2} \otimes L. \quad (5.2.2)$$

By our assumption $w(I_t-, I_t-) = w(-, -)$, which infinitesimally is $w(\delta I-, -) + w(-, \delta I-) = 0 \iff w(\delta I-, -)$ is a symmetric tensor. Since w is a real form of type $(1, 1)$,

$$w(\delta I-, -) = 2i \cdot w(\bar{\tau}-, -) - 2i \cdot w(\tau-, -),$$

is a difference of different types, so the vanishing of the antisymmetric part of $w(\delta I -, -)$ is equivalent to $\iota_\tau w = 0 \in \Lambda^{0,2}$. This implies (5.2.2).

It will be convenient to use the isomorphism $w: T' \rightarrow \Lambda''$ to identify the symmetric tensor $w(\tau -, -) \in \text{Sym}^2(\Lambda^{0,1})$ with a tensor in $\text{Sym}^2(T')$. Locally

$$w = w_{i\bar{j}} dz^i d\bar{z}^j, \tau = \tau_a^b \partial_b \otimes d\bar{z}^a,$$

so

$$w(\tau -, -) = w_{i\bar{j}} dz^i (\tau_a^b \partial_b \otimes d\bar{z}^a) d\bar{z}^j = w_{i\bar{j}} \tau_a^i d\bar{z}^a \otimes d\bar{z}^j = \tau_{\bar{i}}^a w_{a\bar{j}} d\bar{z}^i \otimes d\bar{z}^j.$$

If

$$w(\tau w^{-1}(-), w^{-1}(-)) = G^{ab} \partial_a \otimes \partial_b,$$

for some symmetric G^{ab} , then

$$w(\tau -, -) = G^{ab} w_{a\bar{i}} w_{b\bar{j}} d\bar{z}^i \otimes d\bar{z}^j,$$

hence for all i, j we have $\tau_{\bar{i}}^a w_{a\bar{j}} = G^{ab} w_{a\bar{i}} w_{b\bar{j}}$, or by lifting indexes by the multiplication on the inverse $w^{k\bar{j}}$, we obtain

$$\tau_{\bar{i}}^k = G^{ak} w_{a\bar{i}}. \quad (5.2.3)$$

2. The map $\nabla'_\tau: L \rightarrow \Lambda^{0,1} \otimes L$ is a differential operator of degree ≤ 1 . Denote the sheaf of holomorphic differential operators of degree $\leq k$ acting on L by $\text{Diff}_{\leq k}(L)$. By the definition $\text{Diff}_{\leq}(L) = \mathcal{O}$ is $\mathcal{O}$ -linear operators, while $\phi \in \text{Diff}_{\leq k}(L)$ is determined by the condition $[\phi, f] \in \text{Diff}_{< k}(L)$ for any $f \in \mathcal{O}$. Thus $\text{Diff}(L) = \cup_k \text{Diff}_{\leq k}(L)$ is a sheaf of filtered algebras. We have

$$0 \rightarrow \text{Diff}_{< k}(L) \rightarrow \text{Diff}_{\leq k}(L) \xrightarrow{\sigma} \text{Sym}^k(T') \rightarrow 0,$$

where the projection σ is the principal symbol map. Thus the associated graded to $\text{Diff}(L)$ is the free symmetric algebra $\text{Sym}(T')$ generated by the tangent bundle. Recall that the canonical extension

$$0 \rightarrow \mathcal{O} \rightarrow \text{Diff}_{\leq 1}(L) \rightarrow T' \rightarrow 0,$$

is called the Atiyah class of L , it lies in $\text{Ext}_M^1(T, \mathcal{O}) \simeq H^1(M; \Omega^1)$ and is equal to the class of curvature $[F(\nabla)] \in H^{1,1}$. In our case $[F(\nabla)]$ is proportional to $[iw]$.

By (5.2.2), $\nabla''(\nabla'_\tau) = 0$ and we have a well-defined class $[\nabla'_\tau] \in H^1(M; \text{Diff}_{\leq 1}(L))$. We have a long exact sequence

$$0 \rightarrow H^0(\mathcal{O}) \rightarrow H^0(\text{Diff}_{\leq 1}(L)) \xrightarrow{\sigma} H^0(T) \xrightarrow{F(\nabla)} H^1(\mathcal{O}) \rightarrow H^1(\text{Diff}_{\leq 1}(L)) \xrightarrow{\sigma} H^1(T). \quad (5.2.4)$$

Note that $\sigma(\nabla'_\tau) = \tau$, thus $\sigma([\nabla'_\tau])$ is equal (up to an irrelevant multiplier) to the Kodaira-Spencer class in $H^1(M; T)$ of the deformation δI .

Following Hitchin we interpret the condition on $(s, \delta s)$ in (5.2.1) as a cocycle condition in a certain complex. Fix s such that $\nabla''s = 0$. Consider a complex of $\mathcal{O}$ -modules

$$A(s, \text{Diff}_{\leq 1}(L)) := \text{Cone}(\text{Diff}_{\leq 1}(L) \xrightarrow{s} L)[-1],$$

where the shift ensures that L is in degree 1. The complex $A(s, \text{Diff}_{\leq 1}(L))$ naturally admits a Dolbeault resolution denoted by

$$A^*(\text{Diff}_{\leq 1}, s) = \text{Cone}(\text{Diff}_{\leq 1}(L) \xrightarrow{s} L) \otimes \Lambda^{0,*}.$$

Explicitely we have

$$A^p(\text{Diff}_{\leq 1}, s) = \Lambda^{0,p} \otimes \text{Diff}_{\leq 1}(L) \oplus \Lambda^{0,p-1} \otimes L,$$

where the differential $d_s: A^p \rightarrow A^{p+1}$ is given by the formula

$$d_s(D, w) = (\nabla''(D), \nabla''(w) - (-1)^p D(s)),$$

for all $D \in \Lambda^{0,p} \otimes \text{Diff}_{\leq 1}(L)$ and $w \in \Lambda^{0,p-1} \otimes L$. The (5.2.1) is equivalent to the cocycle condition on

$$(\nabla'_\tau, \delta s) \in A^1(\text{Diff}_{\leq 1}, s).$$

Recall that $M \times B$ is a smooth family of complex structures (M, I_t) such that w is of type $(1, 1)$ for all $I_t, t \in B$. The class $w \in H^1(M, I_t; \Omega^1)$ is well defined. Furthermore, we fixed $U(1)$ -bundle L on M with unitary connection ∇ such that $F(\nabla) = iw$. Then for all t the bundle L admits a natural holomorphic structure with respect to I_t and after passing to sufficiently large power of L , if necessary, one has a complex vector bundle $H^0(M, I_t; L)$ over B . Given tangent vector $\delta t \in T_{t \in B}$ one has the corresponding first order deformation δI of I , i.e.

$$\delta I = 2i(\bar{\tau} - \tau), \quad [\tau] = \text{KS}(\delta t) \in H^1(M, I_t; T),$$

where $\text{KS}: T_{t \in B} \rightarrow H^1(M, I_t; T)$ is the Kodaira-Spencer map. We have the main theorem.

Theorem 5.2.5 (Hitchin). *Suppose that for each $t \in B$, tangent vector $\delta t \in T_{t \in B}$ and a section $s \in H^0(M, I_t; L)$, one has a class*

$$A(s, t, \delta t) \in \mathbb{H}^1(M; A(\text{Diff}_{\leq 1}(L), s)),$$

smoothly depend on $(s, t, \delta t)$ and such that $\sigma(A(s, t, \delta t)) = \text{KS}(\delta t) \in H^1(M, I_t; T)$. Assume the map

$$H^0(M, I_t; T) \xrightarrow{[w]} H^1(M, I_t; \mathcal{O})$$

is an isomorphism, then there is a natural connection on the projective bundle $\mathbb{P}(H^0(M, I_t; L))$.

Proof. We will work at the given point $I = I_t$. By the assumption on w , the long exact sequence (5.2.4) implies that $H^0(\mathcal{O}) \rightarrow H^0(\text{Diff}_{\leq 1}(L))$ is an isomorphism and the symbol map $\sigma: H^1(\text{Diff}_{\leq 1}(L)) \rightarrow H^1(T)$ is an inclusion. The tangent vector δt defines a variation δI equal to $\delta I = 2i(\bar{\tau} - \tau)$ for some d'' -closed $\tau \in \Lambda'' \otimes T'$. Assume $(D, w) \in A^1(\text{Diff}_{\leq 1}, s)$ is a representative of $A(s, t, \delta t)$, for some $D \in \Lambda^{0,1} \otimes \text{Diff}_{\leq 1}(L)$ and $w \in \Lambda^{0,0} \otimes L$ such that $\nabla'' D = 0$ and $Ds + \nabla'' w = 0$. Since

$$\sigma([D]) = \text{KS}(\delta t) = [\tau] = \sigma([\nabla_\tau]) \in H^1(M; T),$$

and $H^1(\text{Diff}_{\leq 1}) \hookrightarrow H^1(T)$ is an inclusion, we have

$$D - \nabla'_\tau = \nabla'' \Phi$$

for some $\Phi \in \Lambda^0 \otimes \text{Diff}_{\leq 1}(L)$. Hence

$$0 = Ds + \nabla'' w = \nabla'_\tau s + \nabla''(\Phi s + w),$$

where we used that $\nabla''(\Phi s) = \nabla''(\Phi)s$ since $\nabla'' s = 0$. Thus setting $\delta s := w + \Phi s$ gives a solution to (5.2.1). Since Φ is unique up to an element from $H^0(\text{Diff}_{\leq 1}(L)) \simeq H^0(\mathcal{O})$, i.e. a constant, we conclude that δs is unique up to a multiple of s . Because of this ambiguity one get a connection on $\mathbb{P}(H^0(M, I_t; L))$. $\square$

3. The map of short exact sequence

$$\begin{array}{ccccccc}
0 & \longrightarrow & \text{Diff}_{\leq 1}(L) & \longrightarrow & \text{Diff}_{\leq 2}(L) & \xrightarrow{\sigma} & \text{Sym}^2(T) \longrightarrow 0 \\
& & \downarrow s & & \downarrow s & & \downarrow \\
0 & \longrightarrow & L & \xrightarrow{\text{id}} & L & \longrightarrow & 0 \longrightarrow 0
\end{array} \tag{5.2.6}$$

gives a short exact sequence of complexes

$$0 \rightarrow \text{Cone}(\text{Diff}_{\leq 1}(L) \xrightarrow{s} L)[-1] \rightarrow \text{Cone}(\text{Diff}_{\leq 2}(L) \xrightarrow{s} L)[-1] \rightarrow \text{Sym}^2(T) \rightarrow 0,$$

and hence, setting $A(\text{Diff}_{\leq k}(L), s) = \text{Cone}(\text{Diff}_{\leq k}(L) \xrightarrow{s} L)[-1]$, by inspecting the induced long exact sequence of hypercohomology, we have a boundary map

$$\Psi: H^0(M; \text{Sym}^2(T)) \rightarrow \mathbb{H}^1(M; A(\text{Diff}_{\leq 1}(L), s)).$$

One can compute $\Psi(G)$ for some $G = G^{ij} \partial_i \otimes \partial_j \in H^0(M; \text{Sym}^2(T))$ using zig-zag lemma applied for the corresponding Dolbeault resolutions. Recall that $\Psi(G)$ is represented in $A^1(s, \text{Diff}_{\leq 1}(L))$ by a lift of G to $A^0(s, \text{Diff}_{\leq 2}(L))$ followed by application of d_s . The second order differential operator $\nabla'_i \circ (G^{ij} \nabla'_j)$ has symbol $G = G^{ij} \partial_i \otimes \partial_j$. Then the class of $[\nabla'', \nabla'_i \circ G^{ij} \nabla'_j]$ is $\Psi(G)$. Then

$$[\nabla'', \nabla'_i \circ G^{ij} \nabla'_j] = -i/2 \cdot (\iota_{\partial_i} w \cdot G^{ij} \nabla'_j + \nabla'_i(G^{ij} \iota_{\partial_j} w))$$

is a first order differential operator with the symbol equal to

$$-i/2 \cdot \iota_{\partial_i} w \cdot G^{ij} \partial_j = -i/2 \cdot w_{i\bar{a}} G^{ij} d\bar{z}^a \cdot \partial_j.$$

Similarly to 5.2.5, a family of complex structures on M parametrized by $t \in B$, defines a symmetric tensor $G(t, \delta t)$ (5.2.3), smoothly dependent on $t \in B$ and $\delta t \in T_{t \in B}$.

Corollary 5.2.7. *Assume the symmetric tensor G is holomorphic, then*

$$\sigma(\Psi(2iG)) = [\tau] \in H^1(M; T).$$

In particular, if $G = (t, \delta t)$ is holomorphic for each $t \in B$ and $\delta t \in T_{t \in B}$, then

$$\Psi(2iG) \in H^1(A^*(s, \text{Diff}_{\leq 1}(L)))$$

satisfies the conditions of Theorem 5.2.5 and provides $\mathbb{P}(H^0(M, I_t; L))$ with a natural projective connection.

References

- [AB83] M. F. Atiyah and R. Bott. “The Yang-Mills equations over Riemann surfaces”. In: *Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences* 308.1505 (1983), pp. 523–615.
- [ADW91] S. Axelrod, S. Della Pietra, and E. Witten. “Geometric quantization of Chern-Simons gauge theory”. In: *Journal of Differential Geometry* 33.3 (1991), pp. 787–902.
- [BK+01] B. Bakalov, A. A. Kirillov, et al. *Lectures on tensor categories and modular functors*. Vol. 21. American Mathematical Society Providence, RI, 2001.

- [BPZ84] A. A. Belavin, A. M. Polyakov, and A. B. Zamolodchikov. “Infinite conformal symmetry in two-dimensional quantum field theory”. In: *Nuclear Physics B* 241.2 (1984), pp. 333–380.
- [Eli+89] S. Elitzur et al. “Remarks on the canonical quantization of the Chern-Simons-Witten theory”. In: *Nuclear Physics B* 326.1 (1989), pp. 108–134.
- [FB04] E. Frenkel and D. Ben-Zvi. *Vertex algebras and algebraic curves*. 88. American Mathematical Soc., 2004.
- [Gin98] V. Ginzburg. “Lectures on D-modules”. With the collaboration of Vladimir Baranovsky and Sam Evens. 1998.
- [Hit87] N. J. Hitchin. “The self-duality equations on a Riemann surface”. In: *Proceedings of the London Mathematical Society* 3.1 (1987), pp. 59–126.
- [Hit90] N. J. Hitchin. “Flat connections and geometric quantization”. In: *Communications in mathematical physics* 131.2 (1990), pp. 347–380.
- [Kac90] V. G. Kac. *Infinite-dimensional Lie algebras*. Vol. 44. Cambridge university press, 1990.
- [Kon88] M. Kontsevich. “Rational conformal field theory and invariants of 3-dimensional manifolds”. 1988.
- [Mne25] P. Mnev. “Lecture notes on conformal field theory”. In: (2025).
- [MS90] G. Moore and N. Seiberg. “Lectures on RCFT”. In: *Physics, geometry and topology*. Springer, 1990, pp. 263–361.
- [Mum70] D. Mumford. *Abelian Varieties*. Vol. 5. Tata Institute of Fundamental Research Studies in Mathematics. Oxford University Press, 1970.
- [NS65] M. S. Narasimhan and C. S. Seshadri. “Stable and unitary vector bundles on a compact Riemann surface”. In: *Annals of Mathematics* 82.3 (1965), pp. 540–567.
- [Seg88] G. B. Segal. “The definition of conformal field theory”. In: *Differential geometrical methods in theoretical physics*. Springer, 1988, pp. 165–171.
- [Wit84] E. Witten. “Non-abelian bosonization in two dimensions”. In: *Communications in Mathematical Physics* 92.4 (1984), pp. 455–472.
- [Wit89] E. Witten. “Quantum field theory and the Jones polynomial”. In: *Communications in mathematical physics* 121.3 (1989), pp. 351–399.